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### Measurement system configured to support installation of a ball and socket joint and method therefor

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#### Abstract

A system is disclosed herein for providing a kinetic assessment and preparation of a prosthetic joint comprising one or more prosthetic components. The system comprises a prosthetic component including sensors and circuitry configured to measure load, position of load on a curved surface, joint stability, range of motion, and impingement. In one embodiment, the system is for a cup and ball joint of a musculoskeletal system. The system further includes a computer having a display configured to graphical display quantitative measurement data to support rapid assimilation of the information. The kinetic assessment measures joint alignment under loading that will be similar to that of a final joint installation. The kinetic assessment can use trial or permanent prosthetic components. Furthermore, adjustments can be made to the applied load magnitude, position of load, and joint alignment by various means to fine-tune an installation.

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#### **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This patent application is a continuation of and claims the benefit of priority to U.S. application Ser. No. 16/592,355, filed on Oct. 3, 2019, which claims the benefit to U.S. Provisional Application Nos. 62/742,179, filed Oct. 5, 2018, and 62/742,207, filed Oct. 5, 2018, of which are incorporated herein by reference in their entireties.

#### **FIELD OF THE INVENTION**

(1) The invention relates in general to medical and surgical devices and more particularly to parameter measurement related to the musculoskeletal system.

#### **BACKGROUND OF THE INVENTION**

(2) The musculoskeletal system of a mammal is subject to breakdown due to many factors such as environment, genetics, diet, usage, and aging. A joint of the musculoskeletal system typically comprises two or more bones that move in relation to one another. Movement is enabled by muscle

tissue and tendons attached to the joint. Ligaments hold and stabilize the one or more joint bones positionally. Cartilage is a wear surface that prevents bone-to-bone contact, distributes load, and lowers friction.

(3) There has been substantial growth in the repair of the human musculoskeletal system. In general, prosthetic orthopedic joints have evolved over time using animal studies, empirical evidence, simulation data, mechanical prototypes, and patient data. The tools being used for orthopedic surgery have been refined over the years but have not changed substantially. Thus, the basic procedure for replacement of an orthopedic joint has been standardized to meet the general needs of a wide distribution of the population. Although the tools, procedure, and artificial joint meet a general need, each replacement procedure is subject to significant variation from patient to patient. The correction of these individual variations relies on the skill of the surgeon to adapt and fit the replacement joint using the available tools to the specific circumstance. It would be of great benefit to provide quantitative measurement data in real-time to support installation of prosthetic components or prosthetic joints.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of a prosthetic shoulder joint in accordance with an example embodiment;
- (2) FIG. 2 is an illustration of a prosthetic shoulder joint in accordance with an example embodiment;
- (3) FIG. 3 is an illustration of a measurement device in a shoulder joint system in accordance with an example embodiment;
- (4) FIG. 4A is a flow diagram for a shoulder joint installation using the measurement device of FIG. 3 in accordance with an example embodiment;
- (5) FIG. 4B is a continuation of the flow diagram from FIG. 4A for the shoulder joint installation using the measurement device of FIG. 3 in accordance with an example embodiment;
- (6) FIG. 5 is an exploded view of the measurement device illustrating components in accordance with an example embodiment;
- (7) FIG. 6 is an exploded side view of the measurement device in accordance with an example embodiment;
- (8) FIG. 7 is an illustration of the upper housing coupled to the bottom housing in accordance with an example embodiment;
- (9) FIG. 8 is a cross-section of the measurement device in accordance with an example embodiment;
- (10) FIG. 9 is an illustration of mechanical features of the enclosure of the measurement device in accordance with an example embodiment;
- (11) FIG. 10 is a cross-sectional view of part of the enclosure illustrating the sensor between the upper housing and the bottom housing in accordance with an example embodiment;
- (12) FIG. 11 is a cross-sectional view of the measurement device illustrating the external curved surface of the upper housing in accordance with an example embodiment;
- (13) FIG. 12 is an illustration of a sensor snap formed in the bottom housing in accordance with an example embodiment;
- (14) FIG. 13 is a cross-sectional view of the bottom housing illustrating a solder hole in accordance with an example embodiment;
- (15) FIG. 14 is an illustration of the support structure in the bottom housing configured to couple to the flexible interconnect in accordance with an example embodiment;
- (16) FIG. 15 is a cross-sectional view of a portion of the upper housing, the bottom housing, and

the humeral tray in accordance with an example embodiment;

(17) FIG. **16** is an illustration of a housing snap on the measurement device to couple the upper housing to the bottom housing in accordance with an example embodiment;

(18) FIG. **17** is an illustration of rigid snaps extending from the bottom housing in accordance with an example embodiment;

(19) FIG. **18** is an illustration of the o-ring in the measurement device in accordance with an example embodiment;

(20) FIG. **19** is an illustration of the bottom housing with the electronic circuitry in accordance with an example embodiment;

(21) FIG. **20** is an illustration of the flexible interconnect in accordance with an example embodiment;

(22) FIG. **21** is an illustration of the measurement device in accordance with an example embodiment;

(23) FIG. **22A** is an illustration of a GUI on the display of the computer in accordance with an example embodiment;

(24) FIG. **22B** is an illustration of the GUI indicating impingement in accordance with an example embodiment;

(25) FIG. **23** is an illustration of the GUI on the display coupled to the computer displaying sensor information related to range of motion from the measurement device in accordance with an example embodiment;

(26) FIG. **24** is an illustration of an option screen in accordance with an example embodiment;

(27) FIG. **25** is an illustration of a range of motion (ROM) overlay on the GUI **380** in accordance with an example embodiment;

(28) FIG. **26** is an illustration of the GUI showing an impingement range of motion assessment in accordance with an example embodiment;

(29) FIG. **27A** is an illustration of measurement data from the measurement device in accordance with an example embodiment;

(30) FIG. **27B** is an illustration of the measurement device transmitting measurement data to the computer and displaying the measurement data on the display in accordance with an example embodiment;

(31) FIG. **28** illustrates a cross-sectional view of the external curved surface of the measurement device as shown in FIG. **21** in accordance with an example embodiment;

(32) FIG. **29A** is an illustration of a spherical coordinate system for calculating force and position in accordance with an example embodiment;

(33) FIG. **29B** is an example of force and position calculations related to sensor location in accordance with an example embodiment;

(34) FIG. **30** is a diagram showing a force magnitude calculation from measurement data from the sensors in accordance with an example embodiment;

(35) FIG. **31** is a diagram showing a position of applied load calculation on the external curved surface of the measurement device using measurement data from the sensors in accordance with an example embodiment;

(36) FIG. **32** is a block diagram of the electronic circuitry in the measurement device in accordance with an example embodiment;

(37) FIG. **33** is a block diagram of the system or computer in accordance with an example embodiment;

(38) FIG. **34** is an illustration of a communication network for measurement and reporting in accordance with an exemplary embodiment;

(39) FIG. **35** is a diagram of a robot supporting installation of a shoulder joint in accordance with an example embodiment;

(40) FIG. **36** is a measurement device in accordance with an example embodiment;

- (41) FIG. 37A is a superior view of the measurement device in accordance with an example embodiment;
- (42) FIG. 37B is a view of the measurement device illustrating the external curved surface in accordance with an example embodiment;
- (43) FIG. 37C is a side view of the measurement device in accordance with an example embodiment;
- (44) FIG. 37D is an anterior view of the measurement device illustrating an under-cut formed in a shim in accordance with an example embodiment;
- (45) FIG. 38 is an exploded view of the measurement device in accordance with an example embodiment;
- (46) FIG. 39 is a view of a cavity of a bottom housing of the measurement device in accordance with an example embodiment;
- (47) FIG. 40 is a cross-sectional view of the measurement device in accordance with an example embodiment;
- (48) FIG. 41A illustrates the measurement device with first shim in accordance with an example embodiment;
- (49) FIG. 41B illustrates the measurement device with a second shim in accordance with an example embodiment;
- (50) FIG. 42 is a cross-sectional view of the external curved surface of the upper housing that is modified to direct loading to predetermined areas of the external curved surface in accordance with an example embodiment;
- (51) FIG. 43 is a block diagram of loading the measurement device in accordance with an example embodiment; and
- (52) FIG. 44 is an illustration of the measurement device illustrating different regions of the external curved surface of the upper housing in accordance with an example embodiment.

#### DETAILED DESCRIPTION OF EMBODIMENTS

- (53) The following description of embodiment(s) is merely illustrative in nature and is in no way intended to limit the invention, its application, or uses.
- (54) For simplicity and clarity of the illustration(s), elements in the figures are not necessarily to scale, are only schematic and are non-limiting, and the same reference numbers in different figures denote the same elements, unless stated otherwise. Additionally, descriptions and details of well-known steps and elements are omitted for simplicity of the description. Notice that once an item is defined in one figure, it may not be discussed or further defined in the following figures.
- (55) The terms “first”, “second”, “third” and the like in the Claims or/and in the Detailed Description are used for distinguishing between similar elements and not necessarily for describing a sequence, either temporally, spatially, in ranking or in any other manner. It is to be understood that the terms so used are interchangeable under appropriate circumstances and that the embodiments described herein are capable of operation in other sequences than described or illustrated herein.
- (56) Processes, techniques, apparatus, and materials as known by one of ordinary skill in the art may not be discussed in detail but are intended to be part of the enabling description where appropriate.
- (57) The orientation of the x, y, and z-axes of rectangular Cartesian coordinates is assumed to be such that the x and y axes define a plane at a given location, and the z-axis is normal to the x-y plane. The axes of rotations about the Cartesian axes of the device are defined as yaw, pitch and roll. With the orientation of the Cartesian coordinates defined in this paragraph, the yaw axis of rotation is the z-axis through body of the device. Pitch changes the orientation of a longitudinal axis of the device. Roll is rotation about the longitudinal axis of the device.
- (58) The orientation of the X, Y, Z axes of rectangular Cartesian coordinates is selected to facilitate graphical display on computer screens having the orientation that the user will be able to relate to

most easily. Therefore the image of the device moves upward on the computer display whenever the device itself moves upward for example away from the surface of the earth. The same applies to movements to the left or right.

(59) Although inertial sensors are provided as enabling examples in the description of embodiments, any tracking device (e.g., a GPS chip, acoustical ranging, accelerometer, magnetometer, gyroscope, inclinometers, MEMs devices) can be used within the scope of the embodiments described.

(60) At least one embodiment is directed to a kinetic orthopedic measurement system to aid a surgeon in determining real time alignment, range of motion, loading, impingement, and contact point of orthopedic implants. Although the system is generic to any orthopedic surgery (e.g., spinal, shoulder, knee, hip, ankle, wrist, finger, toe, bone, musculoskeletal, etc.) the following examples deal with shoulder surgery as a non-limiting example of an embodiment of the invention.

(61) The non-limiting embodiment described herein is related to quantitative measurement based orthopedic surgery and referred to herein as the kinetic system. The kinetic system includes a sensor system that provides quantitative measurement data and feedback that can be provided visually, audibly, or haptically to a surgeon or surgical team. The kinetic system provides the surgeon real-time dynamic data regarding force, pressure, or loading on the shoulder joint, contact and congruency through a full range of motion, and information regarding impingement.

(62) In general, kinetics is the study of the effect of forces upon the motion of a body or system of bodies. Disclosed herein is a system for kinetic assessment of the musculoskeletal system. The kinetic system can be for the installation of prosthetic components or for monitoring and assessment of permanently installed components to the musculoskeletal system. For example, installation of a prosthetic component can require one or more bone surface to be prepared to receive a device or component. The kinetic system is designed to take quantitative measurements of at least the load, position of load, or alignment with the forces being applied to the joint similar to that of a final joint installation. The sensed measurement components are designed to allow ligaments, tissue, and bone to be in place while the quantitative measurement data is taken. This is significant because the bone cuts take into account the kinetic forces where a kinematic assessment and subsequent bone cuts could be substantially changed from an alignment, load, and position of load once the joint is reassembled.

(63) A prosthetic joint installation can benefit from quantitative measurement data in conjunction with subjective feedback of the prosthetic joint to the surgeon. The quantitative measurements can be used to determine adjustments to bone, prosthetic components, or tissue prior to final installation. Permanent sensors can also be housed in final prosthetic components to provide periodic data related to the status of the implant. Data collected intra-operatively and long term can be used to determine parameter ranges for surgical installation and to improve future prosthetic components. The physical parameter or parameters of interest can include, but are not limited to, measurement of alignment, load, force, pressure, position, displacement, density, viscosity, pH, spurious accelerations, color, movement, particulate matter, structural integrity, and localized temperature. Often, several measured parameters are used to make a quantitative assessment. A graphical user interface can support assimilation of measurement data. Parameters can be evaluated relative to orientation, alignment, direction, displacement, or position as well as movement, rotation, or acceleration along an axis or combination of axes by wireless sensing modules or devices positioned on or within a body, instrument, appliance, vehicle, equipment, or other physical system.

(64) While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

(65) The example embodiments shown herein below of the measurement device are illustrative

only and do not limit use for other parts of a body. The measurement device can be a tool, equipment, implant, or prosthesis that measures at least one parameter or supports installation of prosthetic components to the musculoskeletal system. The measurement device can be used on bone, the knee, hip, ankle, spine, shoulder, hand, wrist, foot, fingers, toes, and other areas of the musculoskeletal system. In general, the principles disclosed herein are meant to be adapted for use in all locations of the musculoskeletal system.

(66) At least one embodiment is directed to a system for adjusting or monitoring a contact position of a musculoskeletal joint for stability comprising: a prosthetic component configured to rotate after being coupled to a bone; a sensed prosthesis having an articular surface where the sensed prosthesis is configured to couple to the prosthetic component, where the sensed prosthesis has a plurality of load sensors coupled to the articular surface and a position measurement system configured to measure position, slope, rotation, or trajectory, and a remote system configured to wirelessly receive quantitative measurement data from the sensed prosthesis where the remote system is configured to display the articular surface, where the remote system is configured to display position of applied load to the articular surface, and where the remote system is configured to report impingement as the musculoskeletal joint is moved through a range of motion (ROM).

(67) In general, the joint measurement system disclosed herein is directed to and can be used for any ball and socket joint of a musculoskeletal system. Typically, a first bone terminates in a ball-shaped surface and fits within a second bone having a cup that receives the ball. The first bone is motivated by muscle, tendon, ligament, and tissue to move the first bone such that the ball of the first bone rotates within the cup of the second bone. A ball and socket joint has some of the widest range of motions of different joints within the musculoskeletal system. For example, the shoulder joint and the hip joint are ball and socket joints. The shoulder joint and the hip joint are synovial joints having synovial fluid. The examples disclosed herein below are for a shoulder joint system such as a reverse prosthetic shoulder joint or a standard prosthetic shoulder joint. The reverse prosthetic shoulder and the standard shoulder have a cup and a ball as part of the joint system. The measurement device including the electronic circuitry and sensors can be adapted for either the curved surface of the cup or the curved surface of the ball of the shoulder joint system. All of the shoulder joint examples disclosed herein can be also be used and sized for a hip. The measurement device disclosed herein can be adapted for either a curved surface of the acetabular cup of the hip or it can be adapted for a curved surface of a ball of a femoral head.

(68) FIG. 1 is an illustration of a prosthetic shoulder joint **100** in accordance with an example embodiment. The natural shoulder joint supports a wide range of motion (ROM) when compared to other joints of the musculoskeletal system. The shoulder joint comprises a humerus, a scapula, coracoid process, acromion, and a clavicle. The shoulder joint is encapsulated by a synovial membrane that produces synovial fluid that lubricates the shoulder joint and circulates nutrients to the area. The scapula includes a glenoid cavity. The glenoid cavity has a curved surface. The proximal end of the humerus has a humeral head having a curved surface that couples to the glenoid cavity to support movement and establishes the range of motion of the shoulder joint. A humeral neck extends from the humerus and couples to the humeral head at a predetermined angle. Muscles, tendons and ligaments couple to the humerus, scapula, and clavicle to hold the shoulder joint in place and to move the humerus in relation to the scapula.

(69) Prosthetic shoulder joint **100** comprises a humeral prosthesis **102** and a glenoid prosthesis **114**. Humeral prosthesis **102** includes a stem **104**, a neck **106**, and a head **108**. Head **108** has an exterior curved surface configured to support movement of prosthetic shoulder joint **100**. In one embodiment, the exterior curved surface is convex. The proximal end of a humerus **110** is cut to receive humeral prosthesis **102**. Stem **104** is inserted into the medullary cavity of humerus **110** to couple humeral prosthesis **102** to humerus **110**.

(70) Glenoid prosthesis **114** comprises a glenoid structure **118** and retaining structure **116**. The glenoid cavity on scapula **112** is prepared to receive glenoid prosthesis **114**. Retaining structure **116**

of glenoid prosthesis **114** is configured to couple to scapula **112** to retain glenoid structure **118**. In one embodiment, glenoid structure **118** replaces the glenoid cavity. Glenoid structure **118** has an external curved surface. Glenoid structure **118** is configured to couple to head **108** of humeral prosthesis **102** to support movement of prosthetic shoulder joint **100**. In one embodiment, the external curved surface of glenoid structure is concave. The external curved surface of glenoid structure **118** is low friction to support movement under load by humeral prosthesis **102**.

(71) In one embodiment, glenoid prosthesis **114** is a trialing device that includes a trial measurement device. After measurements have been taken with the trial measurement device the trialing device is removed and a permanent prosthesis is installed. Alternatively, glenoid prosthesis **114** can comprise a tray coupled to retaining structure **116**. A bearing having an articular surface is configured to couple to the tray. In one embodiment, the bearing can be removed from the tray and replaced with the trial measurement device. A further example is a removable humeral head **108** that can be replaced with the trial measurement device. In one embodiment, the trial measurement device on humeral prosthesis **102** can operate separately or in conjunction with the trial measurement device that replaces glenoid structure **118**. In general, the trial measurement device will have at least one sensor configured to measure a parameter. The trial measurement device will have an external curved surface and dimensions similar to glenoid prosthesis **114** or humeral prosthesis **102**. In one embodiment, measurements taken by the trial measurement device will relate to prosthetic shoulder joint **100** range of motion and stability.

(72) FIG. 2 is an illustration of a prosthetic shoulder joint **120** in accordance with an example embodiment. Prosthetic shoulder joint **120** is also known as a reverse prosthetic shoulder joint. Prosthetic shoulder joint **120** comprises a humeral prosthesis **122** and a glenoid prosthesis **130**. Humeral prosthesis **122** comprises a stem **124**, a neck **126**, and a humeral liner **128**. A proximal end of a humerus **110** is cut to receive humeral prosthesis **122**. Stem **124** is inserted into the medullary cavity of humerus **110**.

(73) Glenoid prosthesis **130** comprises a glenoid sphere **132** and a retaining structure **134**. The glenoid cavity of a scapula **112** is prepared for receiving glenoid prosthesis **130**. Retaining structure **134** of glenoid prosthesis **130** is configured to couple to scapula **112** to retain and hold glenoid sphere **132** in a position to couple to humeral prosthesis **122**. In one embodiment, glenoid sphere **132** is configured to couple to a surface of scapula **112** to replace the glenoid cavity. Glenoid sphere **132** has a curved surface configured to couple to humeral liner **128** of humeral prosthesis **122**. Prosthetic shoulder joint **120** is a reverse shoulder because a glenoid sphere that corresponds to a humeral head of the humerus is coupled to the scapula. Also, humeral liner **128** which corresponds to the glenoid cavity of scapula **112** is instead coupled to humerus **110**. Thus, the articulating surfaces have been reversed. In one embodiment, an external curved surface of humeral liner **128** is concave. In one embodiment, the external curved surface of glenoid sphere **132** is convex to couple to the humeral liner **128** and support movement of prosthetic shoulder joint **120**. The external curved surface of humeral liner **128** supports loading and is low friction to support movement of prosthetic shoulder joint **120**.

(74) In one embodiment, humeral liner **128** can be removed and replaced with a trial measurement device. The trial measurement device can be coupled to the neck of humeral prosthesis **122**. For example, neck **126** of humeral prosthesis **122** can terminate in a tray configured to receive humeral liner **128**. Humeral liner **128** is configured to be removable and replaced with the trial measurement device. The trial measurement device will have at least one sensor configured to measure a parameter. The trial measurement device will have an external curved surface and dimensions similar to humeral liner **128**. In one embodiment, measurements taken by the trial measurement device will relate to movement, loading, and stability of humeral prosthesis **122** in prosthetic shoulder joint **130**. In one embodiment, glenoid sphere **132** can be removed and replaced with a second trial measurement device having at least one sensor. In one embodiment, the second trial measurement device can be used instead of the first trial measurement device for assessing

prosthetic shoulder joint **130**. In one embodiment, the first and second trial measurement devices can both be used to provide measurement data for assessing prosthetic shoulder joint **130**. Final prosthetic components are installed after using the first or second trial measurement devices. In one embodiment, one or more of the final prosthetic components can have at least one sensor for measuring a parameter.

(75) FIG. **3** is an illustration of a measurement device **154** in a shoulder joint system **160** in accordance with an example embodiment. In the example embodiment, a reverse shoulder joint is illustrated in the musculoskeletal system. Shoulder joint system **160** comprises computer **162**, glenoid sphere **152**, and humeral prosthesis **158** including measurement device **154**. A glenoid sphere **152** is shown coupled to a prepared bone surface of scapula **140**. The clavicle **142** and coracoid process **144** are shown in relation to the placement of glenoid sphere **152**. In one embodiment, the glenoid cavity of scapula **140** is prepared to receive glenoid sphere **152**. As shown, glenoid sphere **152** couples to a prepared bone surface **146** of scapula **140**. Glenoid sphere **152** can have an anchor or stem to support attachment to scapula **140**. In one embodiment, screws are used to couple glenoid sphere **152** to scapula **140**. Glenoid sphere **152** has an external curved surface configured to couple to measurement device **154**. In the example, glenoid sphere **152** has a convex surface.

(76) In one embodiment, a humeral prosthesis **158** is configured to couple to a humerus **150**. The proximal end of humerus **150** is cut to have a prepared bone surface **148** for receiving humeral prosthesis **158**. The humeral liner has a low friction surface and is configured to support movement of shoulder joint system **160**. The humeral liner is configured to couple humeral tray **156**. In the example, the humeral liner is configured to be removable from humeral prosthesis **158** and is removed in FIG. **3**. Measurement device **154** replaces the humeral liner in humeral tray **156**. In one embodiment, measurement device **154** is configured to be dimensionally equivalent to the humeral liner. In one embodiment, the external curved surface of measurement device **154** is concave and figured to couple to the external curved surface of glenoid sphere **152**. Measurement device **154** comprises at least one sensor and electronic circuitry configured to control a measurement process and transmit measurement data to a computer **162** in proximity to shoulder joint system **160**. In one embodiment, measurement device **154** can further include a position measurement system configured to measure position or movement. The at least one sensor will measure parameters of interest to support installation of shoulder joint system **160**. Typically, computer **162** is in an operating room outside the surgical field where shoulder joint system **160** is being installed. A display **164** includes a graphical user interface (GUI) that supports presenting measurement data in a graphical manner where an operating team can rapidly assimilate the measurement data to verify, adjust or make changes that improve the installation.

(77) In general, at least one component in shoulder joint system **160** has measurement capability. In the example, shoulder joint system **160** is a reverse shoulder system having measurement device **154**. Measurement device **154** can be adapted for use in a standard shoulder joint system comprising a humeral prosthesis and a glenoid prosthesis. Sensors can also be placed in one or both of the humeral prosthesis and glenoid prosthesis of the standard shoulder joint system. Measurement device **154** is not limited to shoulder arthroplasty. Measurement device **154** can be adapted for use for hip, knee, spine, bone, ankle, wrist, fingers, toes, and other parts of the musculoskeletal system.

(78) Quantitative measurement data is needed to kinetically assess and optimize a shoulder joint during surgery. Measurement device **154** delivers quantitative measurement data to a surgeon or surgical team in real-time that support adjustment of the tension on different soft tissues that enable the shoulder and affect range of motion of the shoulder. In one embodiment, measurement device **154** is a temporary or trialing device that is dimensionally substantially equivalent to a corresponding permanent prosthesis. The permanent prosthesis that replaces measurement device **154** in the final prosthetic joint will measure similar to the measurement data provided by

measurement device **154**.

(79) Shoulder joint system **160** is taken through a range of motion (ROM) that is measured by measurement device **154**. For example, a position of humerus **150**, a load magnitude applied to measurement device **154** by glenoid sphere **152**, and a contact point where glenoid sphere **152** couples to measurement device **154** can be measured in real-time through the ROM. Scapular notching is a common complication in a shoulder joint installation. Notching is caused by repetitive contact between humeral prosthesis **158** and the inferior scapular neck that causes an osteolytic reaction which results in polyethylene debris. Adjustments to shoulder joint system **160** can be made when impingement is detected to prevent scapular notching from occurring. The range of motion and loading is monitored to determine whether to adjust tensions on various soft tissue elements enabling the shoulder movement. Adjustments to the soft tissue using the quantitative measurement data can reduce or eliminate impingement, create more stability in the shoulder joint, and increase a range of motion of the shoulder. More specifically, stability is enhanced by using measurements to reduce implant malpositioning, improve subscapularis quality, and adjust muscle tensioning of the shoulder joint. Proper compressive forces of the soft tissue at the glenohumeral joint were found to significantly improve stability in a reverse total shoulder arthroplasty. Moreover, prosthesis designs which lateralize the humerus are inherently more stable because they better tension the rotator cuff and achieve more deltoid wrapping.

(80) FIG. 4A is a flow diagram **170** for a shoulder joint installation using measurement device **154** of FIG. 3 in accordance with an example embodiment. In the example, the measurement device disclosed in flow diagram **170** corresponds to measurement device **154** and is configured to fit into a humeral prosthesis. The measurement device includes at least one sensor and is configured to measure at least one parameter. In one embodiment, the measurement device transmits measurement data to a computer. The measurement data can be reviewed in real-time on a display coupled to the computer. In one embodiment, the measurement device is also compatible with tensioning devices to allow functionality earlier in the surgery such as immediately after a humeral cut. The measurement device supports different size musculoskeletal systems and shoulder joint systems where a single measurement device can support a substantial portion of the population. In one embodiment, the measurement device can be inserted into a humeral trial and implant without special tools. Similarly, the measurement device can be removed from the humeral trial and implant without special tools.

(81) All steps may not be listed, for example steps known in the art that can be used in the method. Also, the steps listed herein do not imply a specific order and may be practiced in different orders depending on the application. In a step **172**, the shoulder is exposed to gain access to the humerus. In a step **174**, the proximal end of the humerus is cut to prepare a humeral side. In one embodiment, the cut is made at a predetermined angle. In one embodiment, a prepared bone surface of the humerus is configured to receive a humeral prosthesis. The stem of the humeral prosthesis is inserted into the medullary canal of the humerus and placed in a predetermined position. The humeral prosthesis may also include a stem protector. In a step **176**, the glenoid prosthesis is prepared for implantation. In one embodiment, one or more bone cuts or bone modifications are made to the scapula for receiving the glenoid prosthesis. In a step **178**, the glenoid prosthesis is inserted and coupled to the scapula. In a step **180**, a trial implant is coupled to the humerus. In a step **182**, a measurement device size is determined for the shoulder application. In one embodiment, different sized measurement devices can be provided for selection. Alternatively, different adapters can be provided to assemble the measurement device for the appropriate size. In a step **184**, the measurement device is activated. In a step **186**, the measurement device is assembled for an appropriate humeral option. The measurement device is configured to be adjusted such that a humeral neck angle of the humeral prosthesis can be changed. For example, different offsets can be chosen such as (0, 2.5, or 5 degrees) to affect a range of motion of the shoulder joint. In a step **188**, a humeral tray and the measurement device are coupled to the humeral stem of the

humeral prosthesis. Thus, the glenoid prosthesis and the humeral prosthesis with the measurement device have been installed respectively to the scapula and the humerus. In a step **190**, the shoulder is reduced. The shoulder is in place with the measurement device sending measurement data to the computer. The display couples to the computer to display the measurement data in real-time to a surgeon or surgical team.

(82) In a step **192**, the shoulder is evaluated through a range of motion (ROM). In one embodiment, the shoulder is moved through one or more predetermined motions to indicate through measurement data, any issues with the prosthetic shoulder joint installation. The measurement device transmits measurement data from the one or more sensors in the measurement device to the computer. The measurement data is displayed in real-time to the surgeon or surgical team. In one embodiment, some of the measurement data is processed by the computer and displayed graphically on the display of the computer to support rapid assimilation of the measurement data by the surgical team. In a step **194**, clinically appropriate adjustments are made based on measurement data from the measurement device.

(83) FIG. **4B** is a continuation of flow diagram **170** from FIG. **4A** for the shoulder joint installation using measurement device **154** of FIG. **3** in accordance with an example embodiment. In a step **196**, the humeral tray trial implant and the measurement device is removed from the humeral prosthesis. In a step **198**, a humeral tray implant is installed onto the humeral prosthesis. In one embodiment, the humeral tray implant is a final implant and not a trialing device. In a step **200**, the insert measurement device is coupled to the humeral prosthesis. In one embodiment, the measurement device is coupled to the humeral tray implant. In one embodiment, the trialing implant is dimensionally similar to the humeral tray implant such that measurement data taken with the measurement device in the humeral tray should be substantially equal to the measurement data taken when the measurement device was in the trial implant. In a step **202**, the shoulder joint is reduced.

(84) In a step **204**, the shoulder is moved through a ROM. The measurement device transmits measurement data to the computer and is displayed on the display. The measurement data is reviewed by the surgeon or surgical team to verify the previous clinically appropriate adjustments using the permanent humeral tray implant coupled to the humeral prosthesis. In one embodiment, further clinically appropriate adjustments can be made to refine or improve the shoulder joint based on the ROM results and quantitative measurement data. In a step **206**, the measurement device is removed from the permanent humeral tray implant. In one embodiment, the measurement device is a disposable device that is disposed of after the surgery is completed. In a step **208**, a humeral liner is coupled to the humeral tray implant. The humeral liner has a curved surface that is configured to couple to the glenoid implant. The humeral liner is dimensionally substantially equal to the measurement device. The humeral liner has a low friction surface that can withstand the loading applied to the shoulder over a range of motion. In a step **210**, the shoulder joint is reduced. In a step **212**, the wound is closed with the permanent shoulder prosthetic components in place. The permanent shoulder will perform equivalently to that measured and adjusted with the measurement device.

(85) FIG. **5** is an exploded view of measurement device **154** illustrating components in accordance with an example embodiment. In general, measurement device **154** generates quantitative measurement data targeted on clinical parameters that effect outcomes, impingement, stability, and range of motion of the joint. In the example, measurement device **154** is configured to be used in a shoulder joint and more specifically in a reverse shoulder joint. Measurement device **154** is configured to measure a functional load range and a maximum load range representative of loading seen in a reverse shoulder joint. Measurement device **154** is configured to detect and report joint forces related to a glenoid sphere coupling to a humeral liner when the shoulder joint is being trialed. Measurement device **154** includes sensors to detect motion and orientation of a humeral prosthesis. Measurement device **154** is configured to detect inferior impingement on a humeral

liner. Measurement device **154** is configured to measure superior impingement on the humeral liner. Measurement device supports an understanding of the soft tissues coupled to the shoulder joint, what soft tissues need to be evaluated and the consequences of adjusting the soft tissue. In one embodiment, the tension of the different soft tissues coupled to the shoulder joint can be evaluated and individually be adjusted within a predetermined range that results in optimal stability. Measurement device **154** is configured to couple to a computer having a display. In one embodiment, the computer and display are within the operating room in view of the surgical team to report the measurement data from measurement device **154**. Typically, the computer and the display are placed outside the sterile field of the operating room. Measurement device **154** includes a low power transceiver to support communications within the operating room but highly attenuated signals outside the operating room. The communication can be secure by encrypting the transmission to prevent the measurement data from being read.

(86) Measurement device **154** and the computer provide load and motion data with minimal lag or delay. In one embodiment, lag or delay is typically less than 2 seconds. In one embodiment, measurement device **154** is designed for a single use and is provided in sterile packaging. In one embodiment, a power source within measurement device **154** has sufficient power for a single use but cannot power a second usage. In one embodiment, measurement device **154** cannot be opened to replace the power source. In one embodiment, a functional life for measurement device **154** is approximately an hour to several hours in a surgical environment. Measurement device **154** comprises biocompatible materials. In one embodiment, measurement device **154** is tested and calibrated before sterile packaging to ensure optimal performance.

(87) Measurement device **154** comprises an upper housing **220** and a bottom housing **222**. Upper housing **220** and bottom housing **222** are configured to couple together to form a hermetically sealed enclosure. Sensors **230**, electronic circuitry **236**, and PC board **234** are hermetically sealed within the enclosure. In the example, upper housing **220** has an exterior curved surface **224** configured to couple to an external curved surface of a glenoid sphere to support movement of the shoulder joint. Upper housing **220** further includes a rim **242** that couples to external curved surface **224**. In one embodiment, upper housing **220** and bottom housing **222** have corresponding retaining features to hold upper housing **220** to bottom housing **222**. Alternatively, screws can be placed through openings **226** to couple upper housing **220** to bottom housing **222**. Bottom housing **222** has a mounting structure **240** configured to couple to a humeral tray of the humeral prosthesis. Glue or an adhesive may also be used to couple upper housing **220** to bottom housing **222**. Mounting structure **240** aligns and retains the enclosure to the humeral prosthesis. Bottom housing **222** includes openings **254** configured to receive structures from upper housing **220**. Openings **254** terminate to a reinforced area that is configured to receive the screws to hold upper housing **220** to bottom housing **222**. A flexible interconnect **228** is configured to couple to printed circuit (PC) board **234**. Flexible interconnect **228** couples sensors **230** to electronic circuitry **236**. In one embodiment, electronic components are mounted to PC board **234**. PC Board **234** includes interconnect to couple the electronic components to form an electronic circuit configured to control a measurement process and transmit measurement data.

(88) In one embodiment, sensors **230** are formed in flexible interconnect **228**. Sensors **230** can be replicated accurately and have similar characteristics when formed at the same time in or on flexible interconnect **228**. A reference sensor **232** can also be formed in or on flexible interconnect **228**. For example, sensors **230** can be load sensors. The load sensors can be an elastic capacitor, a MEMs device, mechanical structure, hydraulic structure, pneumatic structure, strain gauge, a transducer, or a piezo-structure. The load sensors when coupled to a load convert the load to an electrical signal that is provided through flexible interconnect **228** to electronic circuitry **236**. Alternatively, sensors **230** can be discrete sensors that are coupled to flexible interconnect **228**. In the example, sensors **230** as load sensors are elastic capacitors, MEMs devices, or piezo-structures. Sensors **230** couple between upper housing **220** and bottom housing **222**. In one embodiment,

sensors **230** couple to raised regions **238** on an interior surface **244** of bottom housing **222**. Raised regions **238** each have a surface that is not co-planar to interior surface **244** of the bottom housing. In one embodiment, there are an equal number of sensors **230** as raised regions **238**. For example, three load sensors are used to measure loading applied to external curved surface **224**. A load applied to surface **224** of upper housing **226** by the glenoid sphere of the shoulder joint is configured to compress sensors **230**. In one embodiment, screws in openings **226** that couple upper housing **220** to bottom housing **222** can be adjusted to preload sensors **230** thereby placing sensors **230** in a linear operating region for the load range applied to measurement device **154**.

Measurement device **154** can also be placed through a calibration process to ensure optimal performance where each sensor can be measured and correction applied to sensor measurement to ensure linear operation over a load range. The corrections are stored in memory where they can be used to correct each sensor of the system.

(89) FIG. **6** is an exploded side view of measurement device **154** in accordance with an example embodiment. In one embodiment, the side view shows that upper housing **220** couples to bottom housing **222** at a predetermined angle relative to a bottom surface of mounting structure **240**. Electronic circuitry **236**, PC board **234**, flexible interconnect **228**, and sensors **230** are housed in the enclosure formed by upper housing **220** and bottom housing **222**. In one embodiment, electronic circuitry **236** and PC board **234** are placed within a cavity within the interior of bottom housing **222**. Flexible interconnect **228** couples to a connector mounted on PC board **234** to couples sensors **230** to electronic circuitry **236**. An interior surface **256** of upper housing **220** includes raised regions **252**. Raised regions **252** are sensor pads configured to couple to sensors **230**. Raised regions **252** each have a surface that is not co-planar with interior surface **256** of upper housing **220**. In one embodiment, there are an equal number of raised regions **252** as sensors **230**.

(90) Upper housing **220** can be coupled to lower housing **222** by screws, retaining features, adhesive, welding, electrical means, magnetic means or other sealing and fastening methodologies. Upper housing **220** and lower housing **222** can comprise a polymer, ceramic, metal, metal alloy, or material that can support loading of a musculoskeletal joint and provides a low friction surface. In one embodiment, the material comprising upper housing **220** is low friction such that external curved surface **224** is low friction. Alternatively, a low friction coating can be bonded or applied to upper housing **220** to provide a low friction external curved surface **224**. In the example shown herein above, upper housing **220** couples to lower housing **222** by screws. Upper housing **220** includes structures **250** corresponding to openings **254** of FIG. **5**. In one embodiment, structures **250** are cylindrical in shape. A screw placed through an opening **226** couples through a corresponding structure **250**. Structure **250** is a reinforced region of upper housing **220** configured to receive a screw. In one embodiment, structures **250** aligns upper housing **220** to bottom housing **222**. Upper housing **220** aligns to bottom housing **222** such that structures **250** of upper housing **220** couple into openings **254** on surface **244** of bottom housing **222**. In one embodiment, openings **254** terminate to a reinforced region of bottom housing **222**. Screws couple through structures **250** and into the reinforced region in bottom housing **222** to hold upper housing **220** to bottom housing **222**. As mentioned previously, the screws can be adjusted to seal the enclosure and preload sensors **230** for optimal performance.

(91) FIG. **7** is an illustration of upper housing **220** coupled to bottom housing **222** in accordance with an example embodiment. Upper housing **220** is a transparent to show underlying electronic circuitry **236**, flexible interconnect **228**, and sensors **230**. Screws are placed in openings **226** and couple to bottom housing **222** to hold upper housing **220** to bottom housing **222**. In one embodiment, three screws are used to hold upper housing **220** to bottom housing **222**.

(92) Sensors **230** are shown coupling to raised regions **252** of upper housing **220**. Referring briefly to FIG. **6**, raised regions **252** are formed on interior surface **256** of upper housing **220**. Referring briefly to FIG. **5**, sensors **230** also couple to raised regions **238** on interior surface **244** of bottom housing **222**. In one embodiment, raised regions **252** and raised regions **238** have an area greater

than or equal to the area of sensors **230**. Flexible interconnect **228** couples sensors **230** to electronic circuitry **236** within the enclosure. In one embodiment, sensors **230** comprise three sensors configured to measure loading applied to external curved surface **224** of upper housing **220**. In one embodiment, the three sensors are equidistant from one another and located adjacent to rim **242** of upper housing **220** underlying external curved surface **224**. Each sensor **230** is located at a predetermined location on external curved surface **224**. A pressure applied to surface **224** of upper housing **220** compresses sensors **230** between raised regions **252** and raised regions **238** at the predetermined locations. The measurement data from sensors **230** are transmitted from measurement device **154** to computer **162** shown in FIG. 3 within the operating room. The calibration data can be used to adjust the measured output of sensors **230** at measurement device **154** prior to transmission or at computer **162** of FIG. 3. Computer **162** includes a display **164** configured to provide information related to the measurement data. In one embodiment, computer **162** is configured to calculate a load magnitude at a contact point on external curved surface **224** from the measurement data from sensors **230**. In the example, the contact point is an area or region of external curved surface **224** that couples to the glenoid sphere of a reverse shoulder joint. Computer **162** further calculates the position of the contact point on external curved surface **224** from the measurement data from sensors **230** and the locations of sensors on external curved surface **224**. In one embodiment, the measurement data can include data from a position or motion measurement system. The position or motion measurement system is part of electronic circuitry **236**. In one embodiment, the position measurement system comprises one or more inertial sensors. Data from the position measurement system can be used to support the calculations and presentations performed by computer **162** and displayed on display **164**.

(93) Sensors **230** can be tested and calibrated prior to packaging and sterilizing measurement device **154** to further linearize the output. As part of the calibration process the screws can be torqued to different values to pre-load sensors **230**. The pre-loading of sensors **230** can support operation of sensors **230** in a linear region of operation. The calibration data can be stored in memory as part of electronic circuitry **236** and used to correct non-linearities of sensors **230** to provide more accurate measurement data. In the calibration process sensors will be zeroed or measure zero when the external curved surface **224** is unloaded.

(94) FIG. 8 is a cross-section of measurement device **154** in accordance with an example embodiment. A cavity **260** is formed when upper housing **220** couples to bottom housing **222**. Electronic circuitry **236** on printed circuit board **234** is placed in cavity **260**. Cavity **260** includes at least one retaining device to align and retain electronic circuitry **236** and PC board **234**. In one embodiment, sensors **230** are formed in or on flexible interconnect **238**. Flexible interconnect **238** is patterned or formed to place sensors **230** at predetermined locations. As shown, flexible interconnect **238** locates sensor **230** couples between raised region **252** on upper housing **222** and raised region **238** on bottom housing **222**. In one embodiment, a support structure **262** can extend from bottom housing **222** towards upper housing **220** to retain flexible interconnect **238** centrally within measurement device **154**.

(95) FIG. 9 is an illustration of mechanical features of the enclosure of measurement device **154** in accordance with an example embodiment. Measurement device **154** comprises an upper housing **220** and a bottom housing **222** that forms an enclosure for electronic circuitry and one or more sensors. As shown, upper housing **220** comprises a transparent material to show underlying features in bottom housing **222**. In one embodiment, upper housing **220** and bottom housing **222** comprises a biocompatible material. In one embodiment, upper housing **220** and bottom housing **222** can be molded or 3D printed with a polymer material.

(96) Raised regions **238** are formed on interior surface **244** of bottom housing **222**. Raised regions **238** are sensor platforms that extend above interior surface **244** of bottom housing **222**. In one embodiment, each raised region **238** couples to a corresponding sensor. In the example, three raised regions **238** are formed on interior surface **244**. In one embodiment, sensors **230** are capacitors

approximately 4 millimeters in diameter formed in flexible interconnect **228**. In one embodiment, a capacitor can be formed by a first interconnect and a second interconnect separated by a dielectric material within flexible interconnect **228**. In one embodiment, the dielectric material can be polyimide. In one embodiment, the capacitor can be shielded to minimize parasitic coupling of capacitance or signals to the capacitor. In one embodiment, raised regions **238** are greater than or equal to 4 millimeters to support sensors **230**. A sensor snap **270** is a cutout on interior surface **244** of bottom housing **222** in proximity to raised regions **238**. The cutout of sensor snap **270** supports retaining sensor **230** on a corresponding raised region **238**.

(97) A solder hole **272** is a cutout in bottom housing **222** to accommodate interconnect that is used to couple batteries within the enclosure. A flex snap **274** is a retaining feature configured to retain flexible interconnect **228** shown in FIG. **6** to bottom housing **222**. In one embodiment, flex snaps **274** comprises one or more columns that are formed on support structure **262**. Flexible interconnect **228** has one or more openings corresponding to the one or more columns. Flex snaps **274** are pressed through corresponding openings of flexible interconnect **228** to align and retain flexible interconnect **228** to support structure **262**. In one embodiment, flexible interconnect **228** is suspended above a rigid cutout **282** of bottom housing **222**. In one embodiment, rigid cutout **282** is a large cutout area in bottom housing **222**. Electronic circuitry **236** and PC board **234** as shown in FIG. **8** can be placed in rigid cutout **282**. Rigid snap **280** is a retaining feature configured to retain PC board **234** to bottom housing **222**. In one embodiment, rigid snap **280** can be one or more columns configured to couple through one or more openings in PC board **234** to align and retain PC board **234**. In one embodiment, upper housing **220** and bottom housing **222** have retaining features to couple upper housing **220** to bottom housing **222**. In one embodiment, an o-ring is used to hermetically seal the enclosure. In one embodiment, the o-ring circumferentially couples to bottom housing **222**. In one embodiment, the o-ring locks into bottom housing **222**. In one embodiment, a housing snap **278** is one or more male retaining features on an exterior of bottom housing **222**. Housing snaps **278** couples through one or more corresponding openings on upper housing **220** to align and retain upper housing **220** to bottom housing **222**. In one embodiment, the o-ring is made of flexible material and compresses to seal a surface of upper housing **220** and a surface of bottom housing **222** when retained by housing snap **278**. Flexible o-ring compresses and seals the surface of upper housing **220** and the surface of bottom housing **222**. A compressed o-ring applies a force that holds housing snaps **278** within the corresponding openings in upper housing **220**. Tray rim **276** is an extrusion on bottom housing **222** that couples to a humeral tray **156** of a humeral prosthesis **158** as shown in FIG. **3** and transfers loading applied to measurement device **154** to the humeral tray.

(98) FIG. **10** is a cross-sectional view of part of the enclosure illustrating sensor **230** between upper housing **220** and bottom housing **222** in accordance with an example embodiment. In the example, external curved surface **224** of upper housing **220** couples to an external curved surface of glenoid sphere **152** as shown in FIG. **3**. In one embodiment, sensor **230** is an elastic capacitive sensor integrated into flexible interconnect **228** as shown in FIG. **7**. In one embodiment, the elastic capacitive sensor is approximately 0.012 inches thick. Sensor **230** is located adjacent to rim **242** of upper housing **220** underlying external curved surface **224** to maximize an area of measurement. The elastic capacitive sensor is engaged after assembly of upper housing **220** and bottom housing **222**. A gap of 0.010 inches thick is designed into upper housing **220** and bottom housing **222** to pre-load sensor **230** when upper housing **220** and bottom housing **222** are coupled together. In one embodiment, raised region **252** on interior surface **256** of upper housing **220** is flat. Sensor **230** couples to raised region **252**. In one embodiment, raised region **238** on interior surface **244** is also flat. Sensor **230** couples to raised region **238**. In the example, sensor **230** is coupled between raised regions **252** and **238** respectively of upper housing **220** and bottom housing **220** such that sensor **230** is compressed by 0.002 inches when upper housing **220** is coupled to bottom housing **222**. It has been found that the flat surface of raised regions **252** and **238** reduces hysteresis of sensors **230**.

thereby leading to more accurate load measurement. It has been further found that surface roughness affects load measurement. In one embodiment, the flat surfaces are formed smooth to improve measurement consistency.

(99) FIG. 11 is a cross-sectional view of measurement device 154 illustrating external curved surface 224 of upper housing 220 in accordance with an example embodiment. Curved surface 224 is configured to couple to the external curved surface of glenoid sphere 152 as shown in FIG. 3. In one embodiment, sensors 230 comprise three sensors underlying upper housing 220. The three sensors are located underlying external curved surface 224 near rim 242 of upper housing 220. The three sensors are spaced equidistant from one another. In one embodiment, raised regions 252 and 238 respectively on upper housing 220 and bottom housing 222 are located in proximity to rim 242 to locate sensor 230 as high as possible around external curved surface 224 of upper housing 220. Placing sensors 230 near rim 242 maximizes the sensing area that can be measured on external curved surface 224. In one embodiment, the three sensors are placed 44 degrees from a gleno-sphere axis 290. An axis 292 of external curved surface 224 is shown relative to gleno-sphere axis 290. In the example, arrows 294 and 296 are 44 degrees from gleno-sphere axis 290 and indicate a location of sensors 230.

(100) FIG. 12 is an illustration of sensor snap 270 formed in bottom housing 222 in accordance with an example embodiment. Sensor snap 270 comprises cutouts 300 and wings 302. Sensors 230 are coupled to bottom housing 222 in a manner to prevent movement during measurement. Any movement of sensors 230 will introduce error to the measurement data. In the example, movement of the elastic capacitor sensors formed in flexible interconnect 228 creates fluctuations in the capacitive measurements. In one embodiment, wings 302 are formed in flexible interconnect 228 on opposing sides of sensor 230. Cutouts 300 are formed in interior surface 244 of bottom housing 222. Cutouts 300 correspond to wings 302 formed in proximity to sensor 230. Wings 302 press-fit into cutouts 300, to align and retain sensor 230 on raised region 238, and prevent movement of sensor 230 during measurement. Press fitting wings 302 into cutouts 300 have the added benefit of reducing assembly time. As an alternative to cutouts 300, posts can be formed extending from interior surface 244 of bottom housing 222. The posts would be formed on opposing sides of sensor 230 in proximity to sensor 230. The posts would couple to corresponding openings in flexible interconnect 228 to align, retain, and prevent movement of sensor 230 on raised region 238. A further measure to prevent movement would be to glue sensor 230 to raised region 238, glue wings 302 into cutouts 300, or glue both sensor 230 and wings 302 respectively to raised region 238 and cutouts 300.

(101) FIG. 13 is a cross-sectional view of bottom housing 222 illustrating solder hole 272 in accordance with an example embodiment. Solder hole 272 is a cutout in bottom housing 222 for a solder battery connection 310 to couple to PC board 234. As shown, solder hole 272 underlies PC board 234. Solder battery connection 310 couples batteries 312 to a bottom surface of PC board 234 to power electronic circuitry 236. In one embodiment, solder hole 272 allows PC board 234 to lie flat on an interior bottom surface of bottom housing 222. Solder hole 272 positions batteries 312 to allow coupling of flexible interconnect 223 to support structure 262.

(102) FIG. 14 is an illustration of support structure 262 in bottom housing 222 configured to couple to flexible interconnect 228 in accordance with an example embodiment. In one embodiment, support structure 262 extends from an interior bottom surface of bottom housing 222. Support structure 262 is centrally located in rigid cutout 282 as shown in FIG. 9. In one embodiment flexible interconnect 228 is rigidly coupled to support structure 262. Support structure 262 includes flex snaps 274. Flex snaps 274 are columns extending from support structure 262. Flexible interconnect 228 has openings corresponding to flex snaps 274. In one embodiment, flex snaps 274 are oversized or larger than the openings in interconnect 228. The openings of flexible interconnect 228 are pushed on flex snaps 274 which press fits flexible interconnect to support structure 262.

(103) Alternatively, pins can be used to couple flexible interconnect 228 to support structure 262.

Both support structure **262** and flexible interconnect **228** have openings. Pins can be used to couple through the openings in flexible interconnect **228** and into the openings in support structure **262**. In one embodiment, the pins forcibly couple to the openings of support structure **262** as an interference fit to retain flexible interconnect **228** to support structure **262**. An adhesive could also be used to hold the pins to support structure **262**.

(104) In one embodiment, a surface of support structure **262** is not planar or parallel to the bottom surface of bottom housing **222**. In the example, the surface of support structure **262** places flexible interconnect **228** at a 12.5 degree angle relative to the bottom surface of bottom housing **222**. The angle places flexible interconnect **228** in a position to support placement of sensors **230** as shown in FIG. 7. Support structure **262** also places a connector of flexible interconnect **228** in a position to couple to a connector on PC board **234**. In one embodiment, flexible interconnect **228** is suspended above the bottom surface of bottom housing **222**.

(105) FIG. 15 is a cross-sectional view of a portion of upper housing **220**, bottom housing **222**, and humeral tray **156** in accordance with an example embodiment. In the example, bottom housing **222** couples to humeral tray **156** of humeral prosthesis **158**. Humeral tray **156** typically comprises a metal such as stainless steel or titanium. Bottom housing **222** can have a retaining feature that couples to a corresponding retaining feature of humeral tray **156** such that measurement device **154** is retained but removable from humeral tray **156**. In the example, loading on external curved surface **244** of upper housing **220** is transferred through sensors **230** as shown in FIG. 7 to bottom housing **222**. The load distribution to sensors **230** can be unequal depending on the trajectory of the force to external curved surface **244**. The loading couples through bottom housing **222** and is distributed to humeral tray **156**.

(106) Upper housing **220** includes a sidewall **320** configured to couple to a sidewall **328** of bottom housing **222**. Sidewall **320** overlies a portion of sidewall **328** of bottom housing **222** when upper housing **220** is coupled to bottom housing **222**. In one embodiment, bottom housing **222** has an o-ring **322** fitted around a circumferential groove in sidewall **328** of bottom housing **222**. O-ring **322** is configured to hermetically seal the enclosure. In one embodiment, o-ring **322** is compressed when sidewall **320** of upper housing **220** overlies sidewall **328** of bottom housing **222**. As previously mentioned, coupling upper housing **220** to bottom housing **222** pre-loads sensors **230** that corresponds to an external curved surface **244** being unloaded.

(107) Sidewall **328** of bottom housing **222** can have a protrusion **325** extending partially or circumferentially from sidewall **328**. In one embodiment, a first ledge of protrusion **325** couples to sidewall **320** of upper housing **220** from above protrusion **325**. In one embodiment, a second ledge of protrusion **325** couples to a rim **330** of humeral tray **156** from below protrusion **325**. In one embodiment, loading is applied to external curved surface **244**, through load the load sensors, to sidewall **328** of bottom housing **222** to rim **330** and a surface of humeral tray **156** to distribute loading applied to measurement device **154** and to humeral prosthesis **158**. Humeral tray **156** can have an o-ring **332** that aligns, retains, and seals a portion of bottom housing **222** to humeral tray **156**. In one embodiment, bottom housing **222** can have a corresponding groove that accommodates o-ring **332** when measurement device **154** is pressed into humeral tray **156**.

(108) FIG. 16 is an illustration of housing snap **278** on measurement device **154** to couple upper housing **220** to bottom housing **222** in accordance with an example embodiment. Housing snap **278** comprises a protrusion extending from sidewall **328** of bottom housing **222**. In one embodiment, sidewall **320** of upper housing **220** is designed to flex. In one embodiment, sidewall **320** has an opening configured to receive housing snap **278** to retain upper housing **220** to bottom housing **222**. In one embodiment, upper housing **220** slides onto bottom housing **222** such that sidewall **320** of upper housing **220** overlies sidewall **328** of bottom housing **222**. Upper housing **220** and bottom housing **222** are compressed together until the housing snap **278** on sidewall **328** couples through the corresponding opening in sidewall **320** of upper housing **220**. In one embodiment, the housing snap **278** has an angled or sloped wall **340** that facilitates sidewall **320** to flex and slide over the

protrusion until housing snap **278** engages with the corresponding opening. In one embodiment, more than one housing snap **278** is used to retain upper housing **220** to bottom housing **222**. Alternatively, upper housing **220** can be screwed to bottom housing **222** as disclosed in FIGS. 5-7. (109) FIG. **17** is an illustration of rigid snaps **280** extending from bottom housing **222** in accordance with an example embodiment. Rigid snaps **280** are press fit columns extending from the interior bottom surface of bottom housing **222**. In one embodiment, rigid snaps **280** correspond to undersized openings in PC board **234**. In the example, PC board **234** is a multi-layer rigid printed circuit board that interconnects electronic circuitry **236** to form a circuit or system to control a measurement process and transmit measurement from measurement device **154**. Rigid snaps **280** are aligned to openings in PC board **234**. Pressure is applied to PC board **234** until rigid snaps **280** couple through the corresponding openings in PC board **234**. Rigid snaps **280** align and retain PC board **234** within the enclosure. More specifically, rigid snaps **280** prevent movement of PC board **234** while in the shoulder joint. Movement of PC board **234** can introduce movement to leads or flexible interconnect coupling to PC board **234** thereby affecting measurement.

(110) FIG. **18** is an illustration of o-ring **322** in measurement devices **154** in accordance with an example embodiment. Bottom housing **222** has a rim **352**. Sidewall **328** of bottom housing **222** can have a groove **350** around the perimeter to retain an o-ring **322** below rim **352**. In one embodiment, o-ring **322** has a durometer of approximately shore 40. Rim **242** of upper housing **220** overlies o-ring **322** when upper housing **220** couples to bottom housing **222** as shown in FIG. **15**. O-ring **322** hermetically seals the enclosure and prevents the ingress of gas, liquids, or solids from entering measurement device **154**. Load measurement by measurement device **154** is unaffected by o-ring **322**.

(111) FIG. **19** is an illustration of bottom housing **222** with electronic circuitry **236** in accordance with an example embodiment. In one embodiment, PC board **234** is a rigid printed circuit board coupled to the interior bottom surface of bottom housing **222** through rigid snaps **280**. Electronic circuitry **236** comprises electronic components such as processors, digital signal processors, digital logic circuitry, interface circuitry, analog circuitry, buffers, amplifiers, radio frequency circuitry, sensors, passive components, and other circuitry. The electronic components can be mounted to PC board **234**. PC board **234** has multiple layers of interconnect to couple the electronic components to form a circuit that controls a measurement process and transmits measurement data. Flexible interconnect **228** couples sensors **230** to PC board **234**. In one embodiment, flexible interconnect **228** is suspended above PC board **234** and couples to flex plug **362** mounted on PC board **234**. Flex snaps **274** couple through openings in flexible interconnect **228** to retain, align, and prevent movement of flexible interconnect **228**. Flexible interconnect **228** does not have any angled bends because they have been observed to cause sensor anomalies and errant data. In one embodiment, flexible interconnect **228** has a shielding layer **374** to shield sensors **230** and interconnect. Shielding layer **374** is a grounded copper layer that is external to the sensor ground that has showed significant noise improvement. Flexible interconnect **228** has a plug **372** that couples to a flex plug **362** that couples to PC board **234**. Flex plug **362** couples sensors **230** to electronic circuitry **236**. Batteries **312** couple to PC board **234** via an interconnect **310** that underlies PC board **234** in solder hole **312** as shown in FIG. **13**. In one embodiment, batteries **312** are configured to power measurement device **154** for a single operation. An antenna **360** couples to a transceiver circuit on PC board **234**. Antenna **360** transmits measurement data from measurement device **154** within the operating room to computer **162** as shown in FIG. **3** to provide the measurement data for use by a surgeon in real-time. In one embodiment, measurement device **154** is disposed of in an appropriate manner after the shoulder surgery is completed.

(112) FIG. **20** is an illustration of flexible interconnect **228** in accordance with an example embodiment. Flexible interconnect **228** includes sensors **230** configured to measure a force, pressure, or load applied to the measurement device. Sensors **230** can be coupled to flexible interconnect **228** or formed as part of flexible interconnect **228**. In one embodiment, sensors **230**

are capacitors formed in flexible interconnect **228** using two or more layers of interconnect. The capacitors comprise a first metal region and a second metal region separated by a dielectric material. In one embodiment, the capacitors are shielded to prevent noise coupling and to reduce parasitic coupling. In one embodiment, the capacitors comprise more than one capacitor coupled in series or coupled in parallel. In one embodiment, the capacitors can comprise more than one dielectric layer. In general, the capacitors are elastic over a range of operation seen by a shoulder joint. Sensors **230** couple through interconnect in flexible interconnect **228** to a plug **372**. Plug **372** is configured to couple to flex plug **362** on PC board **234** as shown in FIG. **19**. In one embodiment, one or more reference sensors **370** are coupled to or formed in flexible interconnect **228**. Reference sensors **370** are not configured to measure loading applied to the measurement device. Reference sensors **370** are formed identically to sensors **230**. In one embodiment, reference sensors **370** are located near sensors **230** to ensure strong temperature and noise compensation for sensors **230**. If a single reference sensor **370** is used it can be located in an area substantially equidistant from sensors **370**. Alternatively, the single reference sensor **370** can be placed in proximity to one of sensors **370**. In one embodiment, there is a reference sensor **370** for each sensor **230**.

(113) FIG. **21** is an illustration of measurement device **154** in accordance with an example embodiment. In general, a load applied by a prosthetic component coupled to external curved surface **224** will have no path to ground except through the plurality of sensors measuring loading in measurement device **154**. In one embodiment, there will be no parallel paths to apply or dilute loading in measurement device **154**. Measurement device **154** is designed to operate under a maximum load that is applied to external curved surface **224** during installation into a prosthetic joint or when installed in the prosthetic joint. A compression strength of measurement device **154** can accept the maximum load without impacting the load path and accept the maximum load without plastic deformation. In one embodiment, measurement device **154** is configured to withstand torque applied during a humeral reduction. Measurement device **154** has a form factor that matches a target trial implant for the prosthetic joint.

(114) Upper housing **220** is coupled to bottom housing **222** to form the enclosure for housing the power source, electronic circuitry, and one or more sensors. The power source can be a passive storage device, battery, or other means of providing power. Alternatively, power can be wired, inductively or RF coupled to measurement device **154**. The power source has sufficient energy to power the electronic circuitry and sensors for a single joint arthroplasty. Upper housing **220** is retained by one or more housing snap **278** on sidewall **328** of bottom housing **222** coupling through an opening in sidewall **320** of upper housing **220**. Housing snap **278** has an angled or sloped wall **340** that facilitates sidewall **320** of upper housing **220** to flex and slide over housing snap **278** until housing snap **278** couples through the opening in sidewall **320** of upper housing **220** to retain upper housing **220** to bottom housing **222**. In the example, housing snap **278** is configured to keep housing **220** proximate to sensor engagement without applying a load when coupled to housing **222**.

(115) In the example, measurement device **154** couples to the humeral prosthesis. External curved surface **224** couples to a glenoid sphere coupled to a scapula to support movement of a shoulder. In one embodiment, measurement device uses three sensors to measure load and position of load on external curved surface **224**. In one embodiment, at least one reference sensor can also be used to improve the accuracy of the load measurements from each sensor. The interface between the external curved surface **224** and sensors **230** is fully constrained. In one embodiment, the sensing configuration of measurement device **154** uses exactly three load sensors for full constraint if sensors **230** are oriented towards a center of curvature of external curved surface **224** such that all force vectors pass through a same point with no moments to balance. In one embodiment, measurement device **154** measures loading applied to external curved surface **224** in a range of 10-60 lbs. for the shoulder application. The accuracy of the measured load magnitude is 3.5 lbs. or less. In one embodiment, the range and accuracy can be adjusted by changing parameters of the

capacitor sensor such as dielectric thickness or sensor area. The measured capacitance value correlates to loading applied to external curved surface **224**. Alternatively, different sensor types such as a MEMS, strain gauge, or piezo-sensor could be used in place of the capacitor. Measurement device **154** can be operated with a safe overload of 200 percent of the maximum load range. In the example, the position of applied load or contact point of the glenoid sphere on external curved surface **224** has an accuracy of 2 millimeter and 2 degrees or less. This accuracy is given for reference and can be changed or improved depending on the application and requirements of the measurement device. The shoulder joint can be moved through a range of motion and measurement device **154** will provide measurement data in real-time. The measurement data is transmitted to computer **162** in the operating room where computer **162** receives and processes the measurement data and displays the measurement data on display **164** in a form that can be rapidly assimilated by the surgeon and surgical team to support validation or adjustment with quantitative measurement data.

(116) FIG. **22A** is an illustration of a GUI **380** on display **164** of computer **162** in accordance with an example embodiment. Measurement device **154** is in a shoulder joint and transmitting measurement data as shown in FIG. **3**. Components of FIGS. **3**, **7**, **8**, **9**, and **21** may be referred to in the discussion herein below to relate operation of measurement device **154** to what is displayed in GUI **380**. In general, a surgeon moves the shoulder through a free range of motion to generate measurement data to determine stability of the shoulder joint. The measurement data from the sensors in measurement device **154** is transmitted to computer **162**. Computer **162** processes the information and displays the information in a manner where quantitative measurement data can be rapidly assimilated by the surgeon or surgical team. The measurement data can be displayed or it can be presented graphically or audibly.

(117) A picture of a portion of measurement device **154** is displayed on display **164**. In the example, a surface **384** is displayed on display **164** that corresponds to external curved surface **224** as shown in FIG. **21**. In one embodiment, display **164** depicts a curved surface of a ball or socket prosthetic joint system. In one embodiment, radial measurement data from load sensors at predetermined locations as disclosed in FIGS. **29A-31** herein below is used to determine contact point **384** on display **164**. In other words, the movement of contact point **384** is not measured, depicted, or calculated for display **164** as a planar measurement but illustrates movement on a curved surface. Display **164** can further add measurement data or graphics to disclose movement of contact point **384** on external curved surface **224**. In one embodiment, movement of contact point **384** will be non-linear. In one embodiment, display **164** may display a three dimensional type of animation to illustrate to a surgeon or surgical team a location of contact point **384** on external curved surface **224**. This allows the surgeon to understand loading or location on a ball or cup of a prosthetic joint system. Alternatively, more than one view or different orientations of external curved surface **224** can be provided on display **164** to better illustrate a location of contact point **384** on external curved surface **224**. In a reverse shoulder joint, external curved surface **224** couples to a glenoid sphere. A contact point **382** or load centroid location is represented on GUI **380** where the glenoid sphere applies loading to external curved surface **224** of measurement device **154**. Contact point **382** is shown on surface **384** of GUI **380**. A display box **386** discloses the load magnitude in real-time on GUI **380** as the shoulder joint is moved through the range of motion. In one embodiment, computer **162** or measurement device **154** can include software having a force location and load magnitude algorithm that calculates contact point **382** and load magnitude as displayed in display box **386** from the measurement data received from measurement device **154**. In the example, the measurement data comprises information from three sensors measuring loading applied to external curved surface **224**, at least one reference sensor, and a position measurement system. The position measurement system is configured to measure position or motion. In one embodiment, the position measurement system is configured to be housed in measurement device **154**. In one embodiment, the position measurement system is an inertial measurement unit (IMU).

Computer **162** or measurement device **154** can further have quaternion and range of motion algorithms to support measurement of movement and position. Calibration information can be accessed for the IMU or load sensors and used with the force location, load magnitude, and impingement measurements. In one embodiment, calibration information or calibration data corresponds to test measurements on measurement device **154**. In one embodiment, the calibration data can be stored on non-volatile memory such as EEPROM within measurement device **154**. As the shoulder joint is moved through a predetermined range of motion there will be a minimum load magnitude measured at a first location and a maximum load magnitude measured at a second location on surface **384** for the predetermined range of motion. The minimum load magnitude is indicated in display box **388** on GUI **380** that is continuously updated should a lower value occur. Similarly, the maximum load magnitude is indicated in display box **390** on GUI **380** that is continuously updated. In one embodiment, force vector data can be used to detect impingement. GUI **380** will notify the surgeon or surgical team when impingement is detected by audible, visual, or haptic means. In one embodiment, the IMU comprises one or more inertial sensors and is housed in measurement device **154**. The IMU can track position, motion, and can also be used in conjunction with the force vector data or alone to determine impingement.

(118) In the example, an exit button **492**, a LOG button **494**, a zero button **496**, a reset button **392**, and a ROM button **470** are provided on GUI **380**. In one embodiment, exit button **492** toggles between connecting measurement device **154** and disconnecting measurement **154** from computer **162**. In one embodiment, exit button **492** will indicate when measurement device **154** is coupled to computer **162**. In one embodiment, enabling LOG button **494** logs data for 10 seconds. In one embodiment, enabling zero button **496** zeroes load data offsets. In one embodiment, enabling reset button **392** resets display box **388** and display box **390** to the current load magnitude value. In one embodiment, enabling ROM button **370** initiates a range of motion test. ROM button **370** further initializes the IMU for the range of motion test. Battery indicator **526** indicates an amount of power left in the power source. In the example, the power source is one or more batteries and battery indicator **526** indicates the percentage of power remaining in the batteries of measurement device **154** or provide an estimation of an operating time of measurement device **154** based on the average current drain from the batteries. GUI **380** further includes a tracking function that displays dynamic motion of contact point **382** through the full range of motion to evaluate joint kinetics. GUI **380** can also indicate or leave a location trace where loading exceeds a predetermined threshold.

(119) FIG. 22B is an illustration of GUI **380** indicating impingement in accordance with an example embodiment. Components of FIGS. 3, 7, 8, 9, and 21 may be referred to in the discussion herein below to relate operation of measurement device **154** to what is displayed in GUI **380**. GUI **380** shows contact point **382** on surface **384** corresponding to the contact point on surface **224** of measurement device **154**. In the example, contact point **382** is calculated from quantitative measurement data from measurement device **154** and is updated in real-time as the shoulder joint is moved through a range of motion.

(120) Measurement device **154** couples to computer **162** that can be indicated on GUI **380**. An indicator **528** on GUI **380** shows the signal strength of the wireless connection to measurement device **154**. A signal strength is displayed on indicator **528** that provides an indication of the connection and ability to transfer measurement data to computer **162**. In one embodiment, the wireless connection is a Bluetooth low energy connection that opens a connection dialog between computer **162** any Bluetooth devices. Computer **162** is used to select measurement device **154** for connection and initiates the wireless connection. In one embodiment, calibration data and device information from measurement device **154** is downloaded to computer **162**. Measurement device **154** couples to computer **162** and begins to stream measurement data. In one embodiment, GUI **380** zeroes any load data and then begins showing a measured load magnitude at contact point **382** in display box **386**.

(121) In a first step, the ROM button **470** is enabled preparing for measurement device **154** to

measure the shoulder joint as it is moved through a predetermined range of motion. The position measurement system is enabled for measurement. In the example, the position measurement system is an inertial measurement unit. In a second step, the shoulder joint is held motionless at zero degrees adduction for 5 seconds. In a third step, GUI **380** is configured to display a notification to begin movement in abduction. In a fourth step, during the movement, the degrees of rotation and plot will be updating. In a fifth step, at the end of the movement (e.g. full abduction) the user will hold the arm motionless. In a sixth step, measurement data will be captured during the movement and ROM button **470** will indicate that the measurement has ended. In one embodiment, ROM button **470** will change color when the measurement has ended.

(122) Impingement occurs when a prosthetic joint impinges on bone or soft tissue. In the example, scapular notching occurs when the prosthetic shoulder joint impacts bone at some point in the range of motion. Scapular notching typically occurs during an adduction movement. Impingement may also occur in the soft tissue around the prosthetic shoulder joint. Soft tissue impingement is often called acromial impingement. Impingement information can be displayed on GUI **380**. In one embodiment, a rim **520** is used to show if impingement occurs and approximately where the impingement occurs. A portion of rim **520** will be highlighted by a color change or a gray scale change on the portion of rim **520** in proximity to where the impingement occurs. In one embodiment, the portion of rim **520** in proximity to the impingement will turn red when impingement is detected. In one embodiment, a plot **522** is displayed on GUI **380**. Plot **522** shows range of motion angles versus loading for contact point **382**.

(123) FIG. **23** is an illustration of GUI **380** on display **164** coupled to computer **162** displaying sensor information related to range of motion from measurement device **154** in accordance with an example embodiment. In the example, measurement device **154** is in the reverse shoulder joint as shown in FIG. **3**. Components of FIGS. **3**, **7**, **8**, **9**, and **21** may be referred to in the discussion herein below to relate operation of measurement device **154** to what is displayed in GUI **380**. The measurement data comprises data from sensors **230** coupled at predetermined positions in measurement device **154** to measure loading applied to external curved surface **224**. In one embodiment, the predetermined positions of sensors **230** are used to calculate the position of load and the magnitude of load applied to external curved surface **224**. In one embodiment, the measurement data can further comprise information from the position measurement system. In one embodiment, the position measurement system comprises one or more inertial measurement units for tracking position and motion.

(124) Computer **162** converts measurement data from measurement device **154** to a graphical form that a surgeon or surgical team can rapidly assess a status of the shoulder joint. A display box on GUI **380** can be used to provide numerical information related to a parameter measurement. In the example, the shoulder joint can be moved through specific or predetermined range of motions. A motion bar is used to provide information on a predetermined range of motion. The motion bar is a tool of GUI **380** that allows the surgeon to rapidly assess the movement to determine if the shoulder joint functions with known norms or could use further optimization to affect loading or range of motion. Alternatively, a round graphic can be used with a rim and indicator which rotates around it to read for the angle. As shown, four motion bars are displayed on GUI **380**. Each motion bar corresponds to a specific movement of the shoulder joint. In the shoulder example each motion bar will indicate a maximum range of motion for an internal motion and an external motion for the specific movement. As shown, the internal motion maximum is indicated on a left side of the motion bar and a numerical value for the maximum internal motion is listed below the motion bar on the left side. The external motion maximum is indicated on a right side of the motion bar and a numerical value for the maximum internal motion is listed below the motion bar on the right side. A center or zero between internal and external motion is indicated by a bar central to the motion bar. In general, the actual range of motion as the surgeon moves the installed prosthetic joint with measurement device **154** will be less than the internal motion maximum or the external motion

maximum. A first display box will indicate the numerical load magnitude applied to measurement device **154**. A second display box indicates the maximum range of motion (in degrees) achieved by internal movement of the prosthetic joint (by the surgeon) relative to the internal motion maximum. The second display box is placed on the left side of the motion bar and above the motion bar. A third display box indicates the maximum range of motion (in degrees) achieved by external movement of the prosthetic joint (by the surgeon) relative to the external motion maximum. The third display box is placed on the right side of the motion bar and above the motion bar.

(125) The motion bar in GUI **380** graphically displays the same information as the display boxes but in a manner that can be rapidly assimilated to reduce an assessment time. In one embodiment, the surgeon can use the motion bars to determine if the loading and range of motion is within an acceptable range without looking at numerical values. As mentioned previously, the motion bar length indicates a range maximum from internal motion maximum to external motion maximum. The range of motion of the movement by the surgeon of the prosthetic joint can be indicated by a color scale region within the motion bar. The range of motion of the movement of the surgeon can also be indicated by a grey scale region within the motion bar. A color scale can be used in within the motion bar to indicate the magnitude of loading at different points within the range of motion. The color scale can be a load magnitude or correspond to a predetermined load magnitude range over the range of motion. Similarly, the grey in the grey scale region can indicate the magnitude of loading. Each shade of grey can be a load magnitude or correspond to a predetermined load magnitude range. In one embodiment, the surgeon does not need to know the absolute load magnitude at each point over the range of motion but that the load magnitudes are within a predetermined range over the range of motion. The surgeon can “at a glance” determine that the color within the motion bar is correct or that the loading or range of motion is incorrect. In one embodiment, the color or color scale corresponds to an acceptable predetermined range of the load magnitude for the shoulder joint based on clinical evidence that provides optimum performance. In one embodiment, the colors or shades of gray displayed in the motion bar indicate a pattern that the surgeon is looking for over a specific movement of the prosthetic joint. For example, the color scale or grey scale can change as the movement moves to the internal or external maximum. For example, the surgeon can view a grey scale indicating optimal loading over a predetermined range centered between the internal and external movement. Moving outside the predetermined range towards the maximum internal movement or the maximum external movement results in increased or decreased loading that is less than optimal. In general, the surgeon can “at a glance” determine where the optimal loading occurs and at where it is located in the range of motion. The motion bar can also indicate loading or range of motion issues that need to be addressed. For example, adjustments can be made if the loading is non-symmetric about the internal/external movement center, if the optimal loading range does not cover a sufficient range of movement, or if there are loading problems at the extremes. Alternatively, the colors or shades of gray are chosen to allow the surgeon to rapidly assess where the load magnitude is outside the predetermined range and at what range in the movement (e.g. internal motion or external motion) the load magnitude is outside the predetermined range. For example, green can indicate the load magnitude is within a predetermined range. Yellow/Orange can indicate the load magnitude is bordering on being outside the predetermined range. Red can indicate that the load magnitude is higher than acceptable. Blue can indicate that the load magnitude is lower than acceptable. Thus, the surgeon does not need to review numbers but at a glance can determine if the load magnitude over the range of motion is acceptable or needs to be adjusted. The surgeon can then make adjustments such as soft tissue tensioning, modifying a bone surface, changing position of the implant, or shimming the implant to change the load magnitude to be within the predetermined range to name but a few.

(126) Typically, the surgeon is trying to achieve an acceptable range of motion for the internal motion and the external motion of a particular movement of the joint. In one embodiment, the acceptable range of motion can be indicated by dashed lines on the motion bar. A first dashed line

is indicated on the internal motion side of the bar. A second dashed line is indicated on the external motion side of the bar. The surgeon can determine at a glance of the motion bar if the color scale region or the grey scale region overlies the dashed lines or falls short of the acceptable range of motion (defined by the dashed lines) for the internal motion or external motion of the joint movement. Thus, GUI **380** supports rapid assessment of the joint status as it relates to the range of motion and loading over the range of motion. In one embodiment, computer **162** can analyze the measurement data and provide a detailed workflow of the corrections or adjustments to achieve the desired range of motion and loading for the kinetic assessment of the prosthetic joint.

(127) In the example of a shoulder joint, a ROM button **470** is enabled on GUI **380** to initiate a range of motion measurement. Display **164** can be operated via touch screen, remote control, audio control, keyboard, or other device. In the example, GUI **380** shows four motion bars on display **164** after ROM button **470** is enabled. The four motion bars are a motion bar **400**, a motion bar **402**, a motion bar **404**, and a motion bar **406**. Each motion bar has a start/stop button for initiating or stopping a measurement. Although four motion bars are shown in the example, more or less can be displayed depending on the application or joint type. In the example, motion bar **400**, motion bar **402**, motion bar **404**, and motion bar **406** respectively have a start/stop button **430**, a start/stop button **432**, a start/stop button **434**, and a start/stop button **436**. Each motion bar represents a type of movement for a prosthetic shoulder joint that is measured. Motion bar **400** represents a movement comprising internal/external rotation at zero degrees abduction. Motion bar **402** represents a movement comprising internal/external rotation at 45 degrees abduction. Motion bar **404** represents a movement comprising internal/external rotation at zero degrees adduction. Motion bar **406** represents movement of the shoulder joint in extension and flexion.

(128) In the example, start/stop button **432** is enabled to begin a measurement. In one embodiment, all other range of motion tests are disabled when start/stop button **432** is enabled. In one embodiment, starting a new test will reset or redo a finished test. A bar **426** appears across motion bar **402** to indicate position of the shoulder joint within the range of motion of the selected movement. The shoulder joint is moved through the internal/external rotation at 45 degrees abduction. GUI **380** further includes display boxes **414** and **416** in proximity to and above motion bar **402**. GUI **380** also displays a maximum rotation of internal rotation (70 degrees) and the maximum rotation of external rotation (90 degrees) shown respectively at a left end and below motion bar **402** and at a right end and below motion bar **402**. Realistically, maximum rotation or maximum movement both internal and external is not often achievable for joint installations. An acceptable range of movement for the installed shoulder joint is indicated by dashed line **444** corresponding to internal rotation of the shoulder joint at 45 degrees abduction and dashed lines **446** corresponding to external rotation of the shoulder joint at 45 degrees abduction. The actual measured movement range from internal to external corresponds to the color scale region or the gray scale region **462** within motion bar **402**. Note that gray scale region **462** overlies dashed line **444** on the left side of motion bar **402** and overlies dashed line **446** on the right side of motion bar **402**. Gray scale region **462** indicates that the installed prosthetic shoulder joint has an acceptable range of motion for internal/external rotation at 45 degrees abduction. Gray scale region **462** also indicates the loading over the range of motion of the shoulder joint at 45 degrees abduction. The loading applied at the current location is also shown in display box **428**. The gray scale used in gray scale region **462** indicates loading at different points in the range of motion. The surgeon can review gray scale region **462** at a glance to determine if the loading is correct around the movement center between internal and external rotation, if the loading is correct for a sufficient range of motion around the movement center, and behavior or transition of the loading to the internal rotation maximum and the external rotation maximum of the shoulder joint. The surgeon can then perform adjustments to change the loading profile and range of motion indicated by motion bar **402**. As mentioned previously, computer **162** can provide a work flow that provides the adjustments that can be monitored in real-time to produce change in the measurement data related to motion bar

**402** to produce more optimal loading and range of motion.

(129) Motion bar **400**, **404**, and **406** are disabled during the shoulder range of motion measurement with the internal/external rotation at 45 degrees abduction for motion bar **402** of GUI **380**. Motion bar **400** measures a shoulder range of motion having an internal/external rotation at zero degrees abduction when start/stop button **430** is enabled. In the example, the maximum internal rotation is 70 degrees and the maximum external rotation is 80 degrees for motion bar **400**. Dashed lines **440** and **442** respectively indicate an acceptable range of motion for the internal/external rotation at zero degrees abduction. Dashed line **440** couples through motion bar **400** on the left side corresponding to internal rotation. Dashed line **442** couples through motion bar **400** on the right side corresponding to external rotation. The color scale or gray scale region **460** is shown in motion bar **400**. The measured range of motion of the internal rotation at zero degrees adduction is indicated in display box **410** on GUI **380**. Similarly, the measured range of motion of the external rotation at zero degrees abduction is indicated in display box **412**.

(130) Motion bar **404** measures a shoulder range of motion having an internal/external rotation at zero degrees adduction when start/stop button **434** is enabled. In the example, the maximum internal rotation is 70 degrees and the maximum external rotation is 90 degrees for motion bar **404**. Dashed lines **448** and **450** respectively indicate an acceptable range of motion for the internal/external rotation at zero degrees adduction. Dashed line **448** couples through motion bar **404** on the left side corresponding to internal rotation. Dashed line **450** couples through motion bar **404** on the right side corresponding to external rotation. The color scale or gray scale region **464** is shown in motion bar **404**. The measured range of motion of the internal rotation at zero degrees adduction is indicated in display box **418** on GUI **380**. Similarly, the measured range of motion of the external rotation at zero degrees adduction is indicated in display box **420**.

(131) Motion bar **408** measures a shoulder range of motion in extension and flexion when start/stop button **436** is enabled. In the example, the maximum extension is 45 degrees and the maximum flexion is 175 degrees for motion bar **408**. Dashed lines **452** and **454** respectively indicate an acceptable range of motion for extension and flexion of the shoulder joint. Dashed line **452** couples through motion bar **408** on the left side corresponding to the shoulder joint in extension. Dashed line **454** couples through motion bar **408** on the right side corresponding to the shoulder joint in flexion. The color scale or gray scale region **466** is shown in motion bar **408**. The measured range of motion of the shoulder joint in extension is indicated in display box **422** on GUI **380**. Similarly, the measured range of motion of the shoulder joint in flexion is indicated in display box **424**.

(132) FIG. **24** is an illustration of an option screen **482** in accordance with an example embodiment. Referring briefly to FIG. **23**, an options button **480** is pressed on GUI **380** and returns option screen **482**. Option screen **482** allows the user to change a color scale **472** or gray scale region in motion bar **474** that relates color scale **472** or gray scale to measured loading of the joint. Display boxes **476** and **478** respectively indicate a low and a high load value for color scale **472** or gray scale region. In one embodiment, there are four values for setting load range for each color or gray shade. In one embodiment, if a gradient map is selected to indicate loading, a used or acceptable value of loading should be a mid-point of color range **472** or gray scale. In one embodiment, if a solid map is selected to indicate loading, a used value should represent maximum loading for that color range or gray scale. The selected color scale **472** or grey scale range will take effect after exiting options screen **482**. In one embodiment, a redo of the test will result in redrawing motion bar **474** with the new color range or gray scale entered in option screen **384** after enabling return button **470**.

(133) FIG. **25** is an illustration of a range of motion (ROM) overlay **390** on GUI **380** in accordance with an example embodiment. Referring briefly to FIG. **23**, the shoulder joint is taken through four different motions and the measurement data stored. Enabling the ROM display button **490** displays ROM overlay **390**. In the example, GUI **380** graphically displays the movement of contact point **382** to external curved surface **384** on GUI **380** for each of the four different shoulder joint

movements measured in FIG. 23. As previously mentioned, contact point 382 corresponds to the contact point of the glenoid sphere on external curved surface 224 of the shoulder joint. Contact point 382 is calculated from measurement data from load sensors or the IMU in measurement device 154 as shown in FIG. 21. The movement of contact point 382 for a predetermined movement is called a load track. In one embodiment, this is not an active screen or real-time measurement. ROM overlay 390 uses stored measurement data from each of the movements. A load track 500 corresponds to the internal and external movement at zero degrees abduction. A load track 502 corresponds to the internal and external movement at 45 degrees abduction. A load track 504 corresponds to the internal and external movement at zero degrees adduction. A load track 506 corresponds to the movement in extension and flexion of the shoulder joint. Thus, the movement of contact point 382 can be understood for each different movement that is measured and used to determine if a problem may exist in the pattern of movement and the load magnitude at specific points in the movement. In one embodiment, the loading value can be indicated across a load track by color scale or gray scale shade. In one embodiment, computer 162 can analyze the load tracks and provide a workflow to correct or optimize the shoulder joint based on the quantitative measurement data.

(134) FIG. 26 is an illustration of GUI 380 showing an impingement range of motion assessment in accordance with an example embodiment. Enabling I-ROM button 512 produces a graph 514 where trace 510 is continuously active. In one embodiment, the arm and shoulder joint is moved in a “windmill” motion. Trace 510 corresponds to a position of the humerus relative to adduction (graph Y-axis-humerus Z-axis) and horizontal flexion (graph X-axis, humerus Y-axis) is rendered. Track button 516 can be toggled to collect trace data or reset trace for a new data collection. In one embodiment, the measurement ignores the internal/external rotation of the arm. Furthermore, the area covered by trace 510 are the limits of abduction/adduction and horizontal flexion.

(135) FIG. 27A is an illustration of measurement data from measurement device 154 in accordance with an example embodiment. FIG. 27B is an illustration of measurement device 154 transmitting measurement data to computer 162 and displaying the measurement data on display 164 in accordance with an example embodiment. Display 164 includes GUI 380 to support rapid assimilation of measurement data. Housing 220 is made transparent to show placement of electronic circuitry 236 and sensors 530, 532, and 534. Sensors 530, 532, and 534 underlie external curved surface 224 of housing 220. In one embodiment, measurement device 154 shows orientation when placed in a shoulder joint such as a superior position 580, an inferior position 582, an anterior position 586, and a posterior position 584. Sensor seating is configured to orient sensors 530, 532, and 534 toward a center of curvature of external curved surface 224. Sensors 530, 532, and 534 correspond to sensors 230 as shown in FIG. 19 but are individually identified to disclose placement or location within measurement device 154 relative to superior position 580, inferior position 582, anterior position 586, and posterior position 584 in the shoulder joint. Sensors 530, 532, and 534 are respectively labeled S3, S6, and S8 on measurement device 154. Measurement device 154 also has a reference sensor 536 labeled S5. As previously mentioned, sensors 530, 532, and 534 are spaced equidistant from each other and located as close to rim 242 as possible to maximize the measurement area. As shown, sensor 530 is located in proximity to superior position 580. Sensor 532 is located between posterior position 584 and inferior position 582. Sensor 534 is located between anterior position 586 and inferior position 582. Thus, measurement data from each sensor can be correlated to the movement to better understand how shoulder position affects loading. Measurement data from sensors 530, 532, 534, reference sensor 536 is transmitted wirelessly to computer 162. The measurement data from sensors 530, 532, and 534 is used to calculate a load magnitude and a contact point by computer 162 on external curved surface 224 of measurement device 154 of FIG. 21. Display 164 coupled to computer 162 can display the load magnitude and the contact point for viewing by the surgeon and surgical team.

(136) In the example, a shoulder implant is installed in a shoulder of a patient. Measurement device

**154** is inserted in the shoulder joint and powered on. The shoulder is moved through predetermined range of motions. The measurement data from measurement device **154** is captured by computer **162**. In one embodiment, the shoulder can be forced from a neutral position to impingement. Display **164** provides a graph **572** showing load data from sensors **530**, **532**, **534**, reference sensor **536**, and a sum of sensors **530**, **532**, and **534** as the shoulder joint is moved through different predetermined motions. Graph **572** is illustrative of what a surgeon or surgical might see if the measurement data from each sensor was provided graphically. Sensors **530**, **562**, **564**, and **568** are represented by different colors, gray scale shades, or patterned lines on graph **572** as indicated by a legend **574**. In legend **574**, sensor key **560** illustrates measurement data on graph **572** related to sensor **530** (S3). Sensor key **562** illustrates measurement data on graph **572** related to reference sensor **536** (S5) in proximity to superior position **580** of measurement device **154**. Sensor key **564** illustrates measurement data on graph **572** related to sensor **532** (S6) between posterior position **584** and inferior position **582** of measurement device **154**. Sensor key **568** illustrates measurement data on graph **572** related to sensor **534** (S8) between anterior position **586** and inferior position **582** of measurement device **154**. Finally, a sum key **570** illustrates measurement data on graph **572** related to a sum of load measurement data related to sensors **530** (S3), **532** (S6), and **534** (S8).

(137) Box **540** of graph **572** corresponds to a neutral shoulder rotation in adduction. Measurement data from sensors **530** (S3), **532** (S6), and **534** (S8) indicate that sensor **534** is more heavily loaded than sensors **530** (S3) and sensor **532** (S6) during the neutral shoulder rotation in adduction. Reference sensor **536** (S5) is not loaded in the example. The loading on sensor **534** (S8) varies between 5 lbs. to 17 lbs. during the neutral shoulder rotation in adduction. Sum **570** looks similar to sensor **534** (S8) during the neutral shoulder rotation in adduction because the load contributions of sensors **530** (S3) and **532** (S6) are small. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the neutral shoulder rotation in adduction.

(138) Box **542** of graph **572** corresponds to an external shoulder rotation in adduction. Measurement data from sensors **530** (S3), **532** (S6), and **534** (S8) indicate that sensors **530** (S3) and sensor **534** (S8) have a loading less than sensor **532** (S6) during the external shoulder rotation in adduction. Reference sensor **536** (S5) is not loaded in the example. The loading on sensor **532** (S6) varies between 7 lbs. to 15 lbs. during the external shoulder rotation in adduction. Sum **570** looks similar to sensor **532** (S6) during the external shoulder rotation in adduction because the load contributions of sensors **530** (S3) and **535** (S8) are small. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the external shoulder rotation in adduction.

(139) Box **544** of graph **572** corresponds to an internal shoulder rotation in adduction. Measurement data from sensors **530** (S3), **532** (S6), and **534** (S8) indicate that all sensors **530** (S3), sensor **532** (S6), and sensor **534** (S8) have significant loading of greater than 10 lbs. during the internal shoulder rotation in adduction. Reference sensor **536** (S5) is noisy during this measurement having a loading as high as 5 lbs. during the measurement. As previously mentioned, these are just example graphs of measurements. In the initial part of the internal shoulder rotation in adduction, sensor **534** (S8) has a reading of greater than 40 lbs. where it is greatly reduced to under 20 lbs. thereafter. Conversely, the initial part of the internal shoulder rotation in adduction, sensors **530** (S3) and **532** (S6) have a reading of no loading and a loading of greater than 10 lbs. thereafter. The loading on sensor **532** (S6) varies between 7 lbs. to 15 lbs. during the external shoulder rotation in adduction. Sum **570** combines the loading of sensors **530** (S3), **532** (S6), and **534** (S8) during the internal shoulder rotation in adduction which exceeds 50 lbs. in portions of the rotation. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the internal shoulder rotation in adduction.

(140) Box **546** of graph **572** corresponds to a neutral shoulder rotation in abduction. Measurement data from sensors **530** (S3), **532** (S6), and **534** (S8) indicate that sensors **530** (S3) and sensor **534**

(S8) are loaded less than sensor 532 (S6) during the neutral shoulder rotation in abduction. Reference sensor 536 (S5) is unloaded. The loading on sensor 532 (S6) varies between 12 lbs. to greater than 40 lbs. during the neutral shoulder rotation in abduction. Sum 570 looks similar to sensor 532 (S6) during the neutral shoulder rotation in abduction but load components of sensors 530 (S3) and 534 (S8) do contribute such that sum 570 does not overlap sensor key 564 during the neutral shoulder rotation in abduction. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the neutral shoulder rotation in abduction.

(141) Box 548 of graph 572 corresponds to an external shoulder rotation in abduction.

Measurement data from sensors 530 (S3), 532 (S6), and 534 (S8) indicate that sensors 530 (S3) and sensor 534 (S8) are loaded less than sensor 532 (S6) during the external shoulder rotation in abduction. Reference sensor 524 (S5) is unloaded. The loading on sensor 532 (S6) varies between 13-16 lbs. during the external shoulder rotation in abduction. Sum 570 looks similar to load data from sensor 532 (S6) during the external shoulder rotation in abduction but differs because loading from sensors 530 (S3) and 534 (S8) are added. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the external shoulder rotation in abduction.

(142) Box 550 of graph 572 corresponds to an internal shoulder rotation in abduction.

Measurement data from sensors 530 (S3), 532 (S6), and 534 (S8) indicate that sensors 530 (S3) and sensor 534 (S8) are loaded less than sensor 532 (S6) during internal shoulder rotation in abduction. Reference sensor 524 (S5) is unloaded. The loading coupled to sensor 532 (S6) varies between 15-19 lbs. during the internal shoulder rotation in abduction. Sum 570 looks similar to load data from sensor 532 (S6) during the external shoulder rotation in abduction but differs because loading from sensors 530 (S3) and 534 (S8) are added. In general, the surgeon can see graphically where the loading occurs relative to the movement and what each sensor is measuring in the internal shoulder rotation in abduction.

(143) FIG. 28 illustrates a cross-sectional view of external curved surface 224 of measurement device 154 as shown in FIG. 21 in accordance with an example embodiment. In the example, three sensors are used to measure loading and position of load where the glenoid sphere contacts and loads external curved surface 224. The three sensors are equidistant from one another. Sensor 530 and sensor 532 are shown in the cross-sectional view. A center 589 of the glenoid sphere is shown with an axis 588 of the glenoid sphere indicated by a dashed line. Sensor 530 is an angle  $\theta$  from axis 588 of the glenoid sphere. Similarly, sensor 532 is an angle  $\theta$  from axis 588 of the glenoid sphere. In one embodiment, impingement is detected when a measured force angle  $\alpha$  is greater than the angle  $\phi$ . Alternatively, impingement can be detected when the measured force angle  $\alpha$  does not correlate to the assumptions such as the applied force is normal to external curved surface 224, the reaction forces are detected to the center of rotation, there are no moment arms, and simple force balancing applies. In one embodiment, the position measurement system or an IMU can be used to measure a first angle  $\alpha$ . A second angle  $\alpha$  can be measured with the measurement data from the three sensors. The first and second angle  $\alpha$  can be compared to one another as a redundancy check or to determine if the measurement is outside the angle  $\phi$  to determine impingement.

(144) FIG. 29A is an illustration of a spherical coordinate system 600 for calculating force and position in accordance with an example embodiment. In one embodiment, spherical coordinate system 600 is natural and can be used to mathematically describe external curved surface 224 of measurement device 154 as shown in FIG. 21. External curved surface 224 of measurement device 154 couples to a prosthetic component of a joint and is configured to support movement of the joint. In the example, an origin 602 of spherical coordinate system 600 is the center of curvature of external curved surface 224. The positive Z-axis extends from origin 602 to external curved surface 224 intersecting a plane equidistant from each sensor coupled to external curved surface 224. A point in spherical coordinate system 600 can be defined by a radius  $r$ , an angle  $\theta$ , and an angle  $\phi$  as

shown in FIG. 29A. In the example, the radius  $r$  is a radius of curvature of curved surface 224. Theta ( $\theta$ ) is an angle measured from radius  $r$  to the Z-axis. Phi ( $\phi$ ) is an angle measured from the X-axis and dashed line 604. Dashed line 604 can be calculated by the equation  $1\ r \times \sin(\phi)$  on FIG. 29A. Dashed line 604 is on the X-Y plane. Note that an external curved surface that is convex (a ball of the joint instead of the cup) can be modeled similarly.

(145) FIG. 29B is an example of force and position calculations related to sensor location in accordance with an example embodiment. Referring briefly to FIG. 27B, sensors 530, 532, and 534 are configured to measure a force, pressure, or load applied to external curved surface 224 of measurement device 154. As mentioned previously, the sensor locations and the force and position calculations can correspond to either a ball shaped prosthetic component or a cup shaped prosthetic component. In general, sensors 530, 532, and 534 are placed equidistant from each other and have a maximum radius of circle on external curved surface 224 defined by sensors 530, 532, and 534. Referring briefly to FIG. 27B and FIG. 29B sensor 530 is located in proximity to superior position 580 which is called herein as a top sensor of measurement device 154. Sensor 532 is between posterior position 584 and inferior position 582 which is called a bottom left sensor herein. Sensor 534 is between anterior position 586 and inferior position 582 which is called a bottom right sensor herein. In the example, radius  $r$  is given a value of 0.748 inches for a prototype measurement device 154. Sensors 530, 532, and 534 are located as close to the X-Y plane on curved surface 224 as possible to maximize the measurement area on external curved surface 224.

(146) In the example, the radial position of sensors 530, 532, and 534 are described in equation 2 of FIG. 29B. The angle  $\theta_{\text{sub.s}}$  is an angle measured from the Z-axis such that each sensor 530, 532, and 534 is located on dashed circle 606. The angular positions of sensors 530, 532, and 534 are defined in equation 3 of FIG. 29B. The angular position is relative to the X-axis. Sensor 530 is located at  $\phi_{\text{sub.1}} = \pi/2$  on dashed circle 606. Sensor 534 is located at  $\phi_{\text{sub.2}} = 11\pi/6$  on dashed circle 606. Sensor 532 is located at  $\phi_{\text{sub.3}} = 7\pi/6$  on dashed circle 606. In one embodiment, a sensor unit vector represents the direction from the sensor to origin 602. The sensor unit vector is the assumed direction of the sensor reaction force. The equation for the unit vectors related to sensors 530, 532, and 534 are defined by equation 4 where sensors 530, 534, and 532 respectively correspond to unit vector  $\hat{S}_{\text{sub.1}}$ ,  $\hat{S}_{\text{sub.2}}$ , and  $\hat{S}_3$  on FIG. 29B. Sensor 530 has a unit vector  $\hat{S}_{\text{sub.1}} = (0, -0.695, -0.719)$  in polar coordinates which is the top sensor as shown in FIG. 27B. Sensor 534 has a unit vector  $\hat{S}_{\text{sub.2}} = (-0.602, 0.347, 0.719)$  in polar coordinates which is the bottom right sensor as shown in FIG. 27B. Sensor 532 has a unit vector  $\hat{S}_{\text{sub.3}} = (0.602, 0.347, -0.719)$  in polar coordinates which is the bottom left sensor as shown in FIG. 27B.

(147) FIG. 30 is a diagram showing a force magnitude calculation from measurement data from sensors 530, 532, and 534 in accordance with an example embodiment. The force magnitude calculation can be used for a measurement device as a ball or a measurement device as a cup. Thus, measurement device 154 can be implement for a reverse shoulder, regular shoulder, femoral head of a femoral prosthetic component of a hip, or an acetabular cup of a hip. It is assumed in the calculation of the force magnitude and position of applied force that there is no friction on external curved surface 224 of measurement device 154 as shown in FIG. 21. The interface between external curved surface 224 and sensors 530, 532, and 534 as shown in FIG. 27B are fully constrained. In one embodiment, only sensors 530, 532, and 534 are used for full constraint if sensors 530, 532, and 534 are oriented towards the center of curvature of external curved surface 224 such that all force vectors pass through the same point and there are no moments to balance. Reaction force vectors are assumed to be normal to external curved surface 224, and therefore pass through the same point in space at origin 602 that is the center of curvature. In general, sensors 530, 534, and 532 measures loading applied to external curved surface 224 where a scalar reaction force is respectively represented by  $S_{\text{sub.1}}$ ,  $S_{\text{sub.2}}$ , and  $S_{\text{sub.3}}$  for the above listed sensors. Reaction force vectors  $\{\vec{S}\}_{\text{sub.1}}$ ,  $\hat{S}_{\text{sub.2}}$ , and  $\{\vec{S}\}_{\text{sub.3}}$  are given by equation 5 of FIG. 30 where  $i$  is 1, 2, or 3 respectively corresponds to sensors 530 (top

sensor), **534** (bottom right sensor), and **532** (bottom left sensor). As mentioned previously, reaction force vectors  $\{\text{right arrow over (S)}\}.\text{sub.1}$ ,  $\{\text{right arrow over (S)}\}.\text{sub.2}$ , and  $\{\text{right arrow over (S)}\}.\text{sub.3}$  couple through origin **602**. The total reaction force  $\{\text{right arrow over (F)}\}.\text{sub.r}$  is given by equation 6 of FIG. **30** which is the sum of the reaction force vectors  $\{\text{right arrow over (S)}\}.\text{sub.1}$ ,  $\{\text{right arrow over (S)}\}.\text{sub.2}$ , and  $\{\text{right arrow over (S)}\}.\text{sub.3}$ . A balance of forces requires that the applied force vector  $\{\text{right arrow over (F)}\}.\text{sub.a}$  applied to external curved surface **224** is equal to the negative of the total reaction force  $\{\text{right arrow over (F)}\}.\text{sub.r}$  as shown in equation 7 of FIG. **30**. Applied force vector  $\{\text{right arrow over (F)}\}.\text{sub.a}$  couples to a contact point **610** on external curved surface **224** of measurement device **154**. In the reverse shoulder joint example, contact point **610** corresponds to a location where a glenoid sphere applies a force, pressure, or load to measurement device **154** for the measurement data from sensors **530**, **532**, and **534**. Applied force vector  $\{\text{right arrow over (F)}\}.\text{sub.a}$  also couples through origin **602** if the tail is extended. The combined force magnitude can be broken down into X-axis, Y-axis, and Z-axis force components measured by load sensors **530**, **532**, and **534** as shown by equation 8 of FIG. **30** that corresponds to equation 7. The load magnitude applied to external curved surface **224** can be calculated from the measurement data from sensors and more specifically from  $F.\text{sub.ax}$ ,  $F.\text{sub.ay}$ , and  $F.\text{sub.az}$  previously calculated in equation 8 from the scalar reaction forces of sensors **530**, **532**, and **534** broken into X-axis, Y-axis, and Z-axis components and summed. The reported load measured using measurement data from sensors **530**, **532**, and **534** is  $\|\{\text{right arrow over (F)}\}.\text{sub.a}\|$  and can be calculated by equation 9 of FIG. **30** which is the square root of the sum of the squares of  $F.\text{sub.ax}$ ,  $F.\text{sub.ay}$ , and  $F.\text{sub.az}$  calculated. Thus, the load magnitude at contact point **610** can be calculated in real-time from the measurement data from sensors **530**, **532**, and **534**.

(148) FIG. **31** is a diagram showing a position of applied load calculation on external curved surface **224** of measurement device **154** using measurement data from sensors **530**, **532**, and **534** in accordance with an example embodiment. Referring briefly to FIG. **22A**, contact point **382** is indicated on surface **384** of GUI **380** and display **164**. Note that surface **384** is a two-dimensional image of a three dimensional curved surface. Accurate placement of a contact point on a two dimensional image of a three dimensional curved surface is disclosed herein below for GUI **380**. In the example, the point of load application in  $R.\text{sup.3}$  is projected onto the X-Y plane to display load position. In the example, a point **612** ( $x.\text{sub.a}$ ,  $y.\text{sub.a}$ ) is the projected load position on the X-Y plane. In one embodiment, the applied force vector  $\{\text{right arrow over (F)}\}.\text{sub.a}$  is converted to a positional vector  $\{\text{right arrow over (P)}\}.\text{sub.a}$  with the same direction and magnitude of  $r$  as shown in equation 10 of FIG. **31**. Positional vector  $\{\text{right arrow over (P)}\}.\text{sub.a}$  couples to external curved surface **224** at contact point **610** and also couples through origin **602**. The projection of  $\{\text{right arrow over (P)}\}.\text{sub.a}$  onto the X-Y plane is given by equation 11 which is  $(x.\text{sub.a}, y.\text{sub.a}) = (P.\text{sub.ax}, P.\text{sub.ay})$ . Thus, contact point **610** can be accurately represented on a two-dimensional display for view by the surgeon or surgical team in real-time.

(149) In one embodiment, measurement data from force sensors **530**, **532**, and **534** are used to detect impingement corresponding to FIG. **22B**. For example, impingement caused repeatable load spikes peripherally. Load spikes of 10-30 lbs. occur near an impingement point are captured in the measurement data and used to detect impingement. A force magnitude and contact point where a prosthetic component couples external curved surface **224** of measurement device **154** is calculated from measurement data from sensors **530**, **532**, and **534** as disclosed herein above. A position measurement system can also provide measurement data to support the force magnitude and contact point measurement. In the example certain assumptions are made in the calculations. Sensors **530**, **532**, and **534** are placed equidistant from one another at positions that maximize the radius of circle defined by the sensors. Sensors **530**, **532**, and **534** are oriented such that sensor reaction forces are directed to a center of curvature of external curved surface **224**. In one embodiment, it is assumed that no frictional forces or negligible frictional forces occur on external curved surface **224** or at a sensor interface. In one embodiment, reaction force vectors are assumed

to be normal to external curved surface **224** and therefore pass through the center of curvature of external curved surface **224**. There can only be specific combination of forces possible when utilizing one or more of the assumptions disclosed herein above when calculating the load magnitude and the contact point using the forces applied to sensors **530**, **532**, and **534**. In one embodiment, force combinations that are outside the one or more assumptions can mean that the measured force or measured forces by sensors **530**, **532**, and **534** are not normal to external curved surface **224** of measurement device **154**. Alternatively, the force vector may not pass through the center of curvature of external curved surface **224**. Any combination of forces measured by sensors **530**, **532**, and **534** outside the possible combinations determined by one or more assumptions disclosed herein should be reviewed or considered as impingement. Furthermore, time based analysis of the load data from sensors **530**, **532**, and **534** can be used and correlated against known impingement to detect impingement. As mentioned previously, abrupt changes in the measured load vector also equates to impingement. In one embodiment, the position measurement system such as an IMU (inertial measurement unit) can be used to monitor movement and correlate against known joint movement for a particular motion to detect abnormalities that indicate impingement. The joint geometry may also be susceptible to impingement in specific directions. If so, predetermined movements can be performed on the joint to determine if impingement occurs and adjustments made based on quantitative measurement data.

(150) FIG. **32** is a block diagram of electronic circuitry **236** in measurement device **154** in accordance with an example embodiment. In general, electronic circuitry **236** couples to one or more sensors to measure one or more parameters. Components of FIGS. **3**, **5-21** may be referred to in the discussion herein below to relate operation of measurement device **154** to electronic circuitry **236**. The sensors can measure parameters such as, height, length, width, tilt/slope, position, orientation, load magnitude, force, pressure, contact point location, displacement, density, viscosity, pH, light, color, sound, optical, vascular flow, visual recognition, humidity, alignment, position, rotation, inertial sensing, turbidity, bone density, fluid viscosity, strain, angular deformity, vibration, torque, elasticity, motion, acceleration, infection, pain, and temperature to name but a few. In the example, electronic circuitry **236** is configured to control a measurement process, receive measurement data from sensors **230**, receive measurement data from position measurement system **742**, and transmit the measurement data to computer **162** for further analysis and feedback. More specifically, sensors **230** measure a force, pressure, or load at predetermined locations of external curved surface **224**. Sensors **230** at the predetermined locations comprise sensor **530**, **532**, and **534** as shown in FIG. **27B**. Position measurement system **742** measures position, movement, rotation, velocity, acceleration, or distance. In one embodiment, position measurement system comprises an inertial measurement unit (IMU) **744** configured to measure 9 degrees of freedom. IMU **744** can comprise one or more inertial sensors. In one embodiment, sensors **230** and position measurement system **742** is housed in measurement device **154**.

(151) Electronic circuitry **236** comprises power management circuitry **700**, control logic **702**, memory **704**, interface circuitry **706**, position measurement system **742**, and wireless communication circuitry **720**. A power source **740** couples to electronic circuitry **236** to power a measurement process. Power source **740** can be an inductor, super capacitor, storage cell, wired power, wireless power, solar cell, energy harvesting device, or other energy storage medium. In one embodiment, power source **740** comprises batteries **312**. Electronic circuitry **236** further includes a transceiver that can be positioned on or within, or engaged with, or attached or affixed to or within, a wide range of physical systems including, but not limited to instruments, equipment, devices, prosthetic components, or other physical systems for use on or in human bodies and configured for sensing and communicating parameters of interest in real time. Electronic circuitry **236** is coupled together to form an electronic system using multiple layers of interconnect on printed circuit board **234**. Flexible interconnect **228** can be used to couple electronic circuitry **236** to sensors **230** that are remotely located.

(152) Electronic circuitry **236** can be configured to provide two-way communication between measurement device **154** and computer **162**. In one embodiment, measurement device **154** provides quantitative measurement data related to a shoulder joint installation. Measurement device **154** is configured to provide quantitative measurement data related to load magnitude, position of applied load, position, and rotation. The measurement data from measurement device **154** is used by computer **162** in a kinematic assessment to support installation of prosthetic components to ensure optimal loading, balance, stability, alignment, range of motion, and reduce impingement that improves performance and reliability based on clinical evidence.

(153) Power source **740** provides power to electronic circuitry **236** and sensors **230**. The power source **740** can be temporary or permanent. In one embodiment, the power source is not rechargeable. Measurement device **154** is disposable after a single use and the power in batteries **312** are insufficient for a second surgery. Measurement device **154** would be destroyed or disposed of after being used. Alternatively, power source **740** could be rechargeable. Measurement device **154** would be sterilized before being reused. Charging of power source **740** can comprise wired energy transfer or short-distance wireless energy transfer. A charging power source to recharge power source **740** can include, but is not limited to, a battery or batteries, an alternating current power supply, a radio frequency receiver, an electromagnetic induction coil, a photoelectric cell or cells, a thermocouple or thermocouples, or a transducer energy transfer. In one embodiment, energy transfer to power source **740** could be allowed for the single application scenario if power source **740** has insufficient energy to finish the surgery. Furthermore, measurement device **154** can utilize power management circuitry **700** to minimize the power drain of power source **740** while in use or when electronic circuitry **236** is idling.

(154) As previously mentioned, power source **740** in measurement device **154** comprises batteries **312**. Batteries **312** can be recharged by the methods disclosed herein above. Alternatively, power source **740** can be a super capacitor, an inductor, or other energy storage device. An external charging source can be coupled wirelessly to the rechargeable battery, capacitor, or inductive energy storage device through an electromagnetic induction coil by way of inductive charging. The charging operation can be controlled by power management circuitry **700** within electronic circuitry **236**. In one embodiment, power management circuit **700** supports operation of measurement device **154** during charging thereby allowing the surgery to continue if a low charge on power source **740** is detected. For example, power can be transferred to batteries **312**, capacitive energy storage device, or inductive energy storage device by way of efficient step-up and step-down voltage conversion circuitry. This conserves operating power of circuit blocks at a minimum voltage level to support the required level of performance.

(155) Power management circuitry **700** is configured to operate under severe power constraints. In one embodiment, power management circuitry **700** controls power up, power down, and minimizes power usage during operation. The power management circuitry **700** is configured to reduce power dissipation during operation of the system. The power management circuitry **700** can turn off or reduce the power delivered to circuits that are not being used in a specific operation. Similarly, if the system is idle and not being used, the power management circuitry **700** can put other unused circuitry in a sleep mode that awakens prior to the next measurement being made. Power management circuitry **700** can include one or more voltage regulation circuits that provide a plurality of different stable voltages to electronic circuitry **236** and sensors **230**.

(156) In one configuration, a charging operation of power source **740** can further serve to communicate downlink data to electronic circuitry. For instance, downlink control data can be modulated onto the energy source signal and thereafter demodulated from an inductor in electronic circuitry **230**. This can serve as a more efficient way for receiving downlink data instead of configuring an internal transceiver within electronic circuitry **230** for both uplink and downlink operation. As one example, downlink data can include updated control parameters that measurement device **154** uses when making a measurement, such as external positional information

or for recalibration purposes. It can also be used to download a serial number or other identification data.

(157) Control logic **702** controls a measurement process or sequence that engages the sensors, converts the measurement data into a useable format, and transmits the information. Control logic **702** can comprise digital circuitry, a microcontroller, a microprocessor, an ASIC (Application Specific Integrated Circuit), a DSP (Digital Signal Processing), a gate array implementation, a standard cell implementation, and other circuitry. Control logic **702** couples to memory **704**. Memory **704** is configured to store measurement data, software routines, diagnostics/test routines, calibration data, calibration algorithms, workflows, and other information or programs. In one embodiment, one or more sensors may be continuously enabled and sampled periodically to control logic **702**. Control logic **702** controls the measurement process, stores the measurement data in memory, or transmit the measurement data in real-time. Control logic **702** can include dedicated ports that couple to a sensor to continuously receive measurement data or receive updated measurements at high sample rates. Alternatively, control logic **702** can select a sensor to be measured. For example, multiple sensors can be coupled to control logic **702** via a multiplexer. Control logic **702** controls which sensor is coupled through the multiplexer to sample and output the measurement data. Multiplexed measurement data works well when the measurement data is not critical or can be sampled occasionally as needed. Control logic **702** can also select and receive measurement data from different sensors in a sequence or simultaneously through parallel channels. Control logic **702** can be configured to monitor the measurement data from a sensor but transmit measurement data only when a change occurs in the measurement data. Furthermore, control logic **702** can modify the measurement data prior to transmitting the measurement data to computer **162**. For example, the measurement data can be corrected for non-linearity using calibration data. In one embodiment, a microcontroller with Bluetooth low energy (BLE) is used with an analog to digital converter to convert analog values to digital.

(158) Interface circuitry **706** couples between sensors **230** and control logic **702**. Interface circuitry **706** supports conversion of a sensor output to a form that can be received by computer **162**. Interface circuitry **706** comprises digital circuitry and analog circuitry. The analog circuitry can include multiplexers, amplifiers, buffers, comparators, filters, passive components, analog to digital converters, and digital to analog converters to name but a few. In one embodiment interface circuitry **706** uses one or more multiplexers to select a sensor for providing measurement data to control logic **702**. Control logic **702** is configured to provide control signals that enable the multiplexer to select the sensor for measurement. The multiplexer can be enabled to deliver the measurement data to control logic **702**, memory **704**, or to be transmitted. Typically, at least one analog to digital conversion or digital to analog conversion of the measurement data occurs via the interface circuitry **706**.

(159) Sensors **230** couple through interface circuitry **706** to control logic **702**. Alternatively, interface circuitry **706** can couple directly to circuitry for transmitting measurement data as it is measured. The physical parameter or parameters of interest measured by sensors **230** are force, pressure, or load as disclosed herein but sensors **230** can further include other sensors that measure height, length, width, tilt/slope, position, orientation, load magnitude, force, pressure, contact point location, displacement, density, viscosity, pH, light, color, sound, optical, vascular flow, visual recognition, humidity, alignment, rotation, inertial sensing, turbidity, bone density, fluid viscosity, strain, angular deformity, vibration, torque, elasticity, motion, and temperature. Often, a measured parameter is used in conjunction with another measured parameter to make a kinetic and qualitative assessment. In joint reconstruction, portions of the musculoskeletal system are prepared to receive prosthetic components. Preparation includes bone cuts or bone shaping to mate with one or more prosthesis. Parameters can be evaluated relative to orientation, stability, alignment, impingement, direction, displacement, or position as well as movement, rotation, or acceleration along an axis or combination of axes by wireless sensing modules or devices positioned on or within a body,

instrument, appliance, vehicle, equipment, or other physical system.

(160) Sensors **230** can directly or indirectly measure a parameter of interest. For example, a load sensor in measurement device **154** can comprise a capacitor, a piezo-sensor, or a MEMS sensor that can compress as loading is applied to the load sensor. Measuring load with a capacitor is an indirect form of sensing as the capacitance value of the capacitor will change with the amount of loading applied to the capacitor. The capacitive measurement data can be sent to computer **162** for further processing. Computer **162** can include software and calibration data related to the elastic capacitors. The load measurement data can be converted from capacitance values to load measurements. Computer **162** can store calibration data that can be used to curve fit and compensate for non-linear output of a sensor over a range of operation. Furthermore, the individual sensor measurement can be combined to produce other measurement data by computer **162**. In keeping with the example of load measurement data, the individual load measurement data can be combined or assessed to determine a location where the load is applied to a surface to which the load sensors couple. The measurement data can be displayed on a display that supports a surgeon rapidly assimilating the measurement data. For example, the calculated measurement data on the location of applied load to a surface may have little or no meaning to a surgeon. Conversely, an image of the surface being loaded with a contact point displayed on the surface can be rapidly assimilated by the surgeon to determine if there is an issue with the contact point.

(161) In one embodiment, shoulder joint system **160** transmits and receives information wirelessly. Wireless operation reduces clutter within the surgical area, wired distortion of, or limitations on, measurements caused by the potential for physical interference by, or limitations imposed by, cables connecting a device with an internal power with data collection, storage, or display equipment in an operating room environment. Electronic circuitry **236** includes wireless communication circuitry **720**. In one embodiment, wireless communication circuitry **720** is configured for short range telemetry and battery operation. Typically, measurement device **154**, and computer **162** are located in an operating room such that the transmission of measurement data from measurement device **156** to computer **162** is less than 10 meters. As illustrated, the exemplary communications system comprises wireless communication circuitry **720** of measurement device **154** and receiving system wireless communication circuitry **722** of computer **162**. Wireless communications circuitry **720** comprises, but is not limited to, the antenna **360**, a matching network **716**, the telemetry transceiver **714**, a CRC circuit **712**, a data packetizer **710**, and a data input **708**. Wireless communication circuitry **720** can include more or less than the number of components shown and are not limited to those shown or the order of the components.

(162) Similarly, computer **162** includes wireless communication circuitry **722**. Wireless communication circuitry **722** comprises an antenna **724**, a matching network **726**, a telemetry receiver **728**, a CRC circuit **730**, and a data packetizer **732**. Notably, other interface systems can be directly coupled to the data packetizer **732** for processing and rendering sensor data. In general, electronic circuitry **236** couples to sensors **230** and is configured to transmit quantitative measurement data to computer **162** in real-time to process, display, analyze, and provide feedback. Measurement device **154** includes a plurality of load sensors configured to measure loads applied to external curved surface **224**. Measurement device **154** further includes an inertial measurement unit comprising one or more inertial sensors and other parameter measurement sensors as listed herein above. The measurement data from the plurality of load sensors and the inertial sensors is transmitted to computer **162**. Computer **162** can calculate a load magnitude applied to external curved surface **224** from the plurality of load sensors. In the example, three load sensors are used for the measurement. Computer **162** can further calculate a position of applied load (contact point) to external curved surface **224** of measurement device **154**. Measurement device **154** can further use measurement data from position measurement system **742** to monitor position and movement of measurement device **154** or a prosthetic component. The position or tracking data from position measurement system **742** is also sent to computer **162**. The results can also be displayed on display

**164** of computer **162**. In one embodiment, measurement data from position measurement system **742** can be used to measure range of motion, alignment, and impingement. In one embodiment, the transmission of the measurement data from different sensors or components can be sent on different channels or the measurement data can be sent at different times on the same channel. (163) As mentioned previously, wireless communication circuitry comprises data input **708**, data packetizer **710**, CRC circuit **712** telemetry transmitter **714**, matching network **716**, and antenna **718**. In general, measurement data from sensors **230** is provided to data input **708** of wireless communication circuitry **720**. In one embodiment, the measurement data from sensors **230** can come directly from interface circuitry **706**, from memory **704**, from control logic **702**, or from a combination of paths to data input **708**. In one embodiment, measurement data can be stored in memory **704** prior to being provided to data input **708**. Data packetizer **710** assembles the measurement data into packets and includes sensor information received or processed by control logic **702**. Control logic **702** can comprise specific modules for efficiently performing core signal processing functions of the measurement device **154**. Control logic **702** provides the further benefit of reducing the form factor to meet dimensional requirements for integration into measurement device **154**.

(164) In general, measurement data from measurement device **154** is encrypted. In one embodiment, the output of data packetizer **710** couples to the input of CRC circuit **712**. CRC circuit **712** applies error code detection on the packet data. The cyclic redundancy check is based on an algorithm that computes a checksum for a data stream or packet of any length. These checksums can be used to detect interference or accidental alteration of data during transmission. Cyclic redundancy checks are especially good at detecting errors caused by electrical noise and therefore enable robust protection against improper processing of corrupted data in environments having high levels of electromagnetic activity. The output of CRC circuit **712** couples to the input of telemetry transceiver **714**. The telemetry transceiver **714** then transmits the CRC encoded data packet through the matching network **716** by way of the antenna **360**. Telemetry transceiver **714** can increase a carrier frequency in one or more steps and add the information or measurement data from measurement device **154** to the carrier frequency. The matching network **716** provides an impedance match for achieving optimal communication power efficiency between telemetry transmitter **714** and antenna **360**.

(165) Antenna **360** can be integrated with components of the measurement device **154** to provide the radio frequency transmission. The substrate for the antenna **360** and electrical connections with the electronic circuitry **236** can further include the matching network **716**. In one embodiment, the antenna **360** and a portion of the matching network **716** can be a wire or formed in printed circuit board **234** that interconnects the components that comprise electronic circuitry **236**. This level of integration of the antenna and electronics enables reductions in the size and cost of wireless equipment. Potential applications may include, but are not limited to any type musculoskeletal equipment or prosthetic components where a compact antenna can be used. This includes disposable modules or devices as well as reusable modules or devices and modules or devices for long-term use.

(166) The process for receiving wireless communication circuitry **722** is the opposite of the sending process. Antenna **724** receives transmitted measurement data from wireless communication circuitry **720**. Wireless communication circuitry **720** can transmit at low power such that receiving wireless communication circuitry **722** must be in proximity, for example within 10 meters to receive measurement data. Antenna **724** couples to matching network **726** that efficiently couples the measurement data to telemetry transmitter circuit **728**. The measurement data can be sent on a carrier signal that supports wireless transmission. The measurement data is stripped off from the carrier signal by telemetry transmitter **728**. The measurement data is received by CRC circuit **730** from telemetry transmitter **728**. CRC circuit **730** performs a cyclic redundancy check algorithm to verify that the measurement data has not been corrupted during transmission. The CRC circuit **730**

provides the checked measurement data to data packetizer **732**. Data packetizer **732** reassembles the measurement data where it is provided to USB interface **734**. USB interface **734** provides the measurement data to computer **162** for further processing.

(167) It should be noted that the measuring, transmitting, receiving, and processing of the measurement data can be performed in real-time for use by a surgeon to support installation of a shoulder joint. In one embodiment, computer **162** displays at least a portion of one prosthetic component. In the example, external curved surface **224** and rim **242** of measurement device **154** is displayed on display **164** coupled to computer **162**. Measurement data from sensors **230** and position measurement system **742** is used to calculate a load magnitude and a position of applied load on external curved surface **224** of measurement device **154**. The location of each load sensor is known relative to external curved surface **224**. The position of applied load can be calculated using the location information from each load sensor and the load magnitude at each location by computer **162** as disclosed in detail herein above. The position of applied load is also called contact point **382** on GUI **380** of display **164**. Similarly, the load magnitude at contact point **382** can be calculated from the three load sensors and the three load sensor locations. Typically, the shoulder joint is moved through a predetermine range of motion. The minimum load, the maximum load, and the load at the current location is displayed on GUI **380** respectively in display boxes **380**, **390**, and **386**. The amount of rotation or range of motion can also be indicated. These measurements are measured or calculated in real-time. Rim **242** can also be highlighted to indicate impingement during the predetermined range of motion. In one embodiment, rim **242** will high light an area of rim **242** in proximity to the measured impingement. Adjustments can be performed that affect alignment, loading, position of load, rotation, or other parameters and monitored in real-time on display **164**. The adjustments can support optimization after the measured parameters are within specification to fine tune the prosthetic component installation with quantitative measurement data.

(168) FIG. **33** is a block diagram of the system or computer in accordance with an example embodiment. The exemplary diagrammatic representation of a machine, system, or computer in the form of a system **800** within which a set of instructions, when executed, may cause the machine to perform any one or more of the methodologies discussed above. In some embodiments, the machine operates as a standalone device. In some embodiments, the machine may be connected (e.g., using a network) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client user machine in server-client user network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

(169) The machine may comprise a server computer, a client user computer, a personal computer (PC), a tablet PC, a laptop computer, a desktop computer, a control system, logic circuitry, a sensor system, an ASIC, an integrated circuit, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. It will be understood that a device of the present disclosure includes broadly any electronic device that provides voice, video or data communication. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(170) System **800** may include a processor **802** (e.g., a central processing unit (CPU), a graphics processing unit (GPU, or both), a main memory **804** and a static memory **806**, which communicate with each other via a bus **808**. System **800** may further include a video display unit **810** (e.g., a liquid crystal display (LCD), a flat panel, a solid state display, or a cathode ray tube (CRT)). System **800** may include an input device **812** (e.g., a keyboard), a cursor control device **814** (e.g., a mouse), a disk drive unit **816**, a signal generation device **818** (e.g., a speaker or remote control) and a network interface device **820**.

(171) The disk drive unit **816** can be other types of memory such as flash memory and may include a machine-readable medium **822** on which is stored one or more sets of instructions **824** (e.g.,

software) embodying any one or more of the methodologies or functions described herein, including those methods illustrated above. Instructions **824** may also reside, completely or at least partially, within the main memory **804**, the static memory **806**, and/or within the processor **802** during execution thereof by the system **800**. Main memory **804** and the processor **802** also may constitute machine-readable media.

(172) Dedicated hardware implementations including, but not limited to, application specific integrated circuits, programmable logic arrays and other hardware devices can likewise be constructed to implement the methods described herein. Applications that may include the apparatus and systems of various embodiments broadly include a variety of electronic and computer systems. Some embodiments implement functions in two or more specific interconnected hardware modules or devices with related control and data signals communicated between and through the modules, or as portions of an application-specific integrated circuit. Thus, the example system is applicable to software, firmware, and hardware implementations.

(173) In accordance with various embodiments of the present disclosure, the methods described herein are intended for operation as software programs running on a computer processor. Furthermore, software implementations can include, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing can also be constructed to implement the methods described herein.

(174) The present disclosure contemplates a machine readable medium containing instructions **824**, or that which receives and executes instructions **824** from a propagated signal so that a device connected to a network environment **820** can send or receive voice, video or data, and to communicate over the network **826** using the instructions **824**. The instructions **824** may further be transmitted or received over the network **826** via the network interface device **820**.

(175) While the machine-readable medium **822** is shown in an example embodiment to be a single medium, the term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable medium” shall also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure.

(176) The term “machine-readable medium” shall accordingly be taken to include, but not be limited to: solid-state memories such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other re-writable (volatile) memories; magneto-optical or optical medium such as a disk or tape; and carrier wave signals such as a signal embodying computer instructions in a transmission medium; and/or a digital file attachment to e-mail or other self-contained information archive or set of archives is considered a distribution medium equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a machine-readable medium or a distribution medium, as listed herein and including art-recognized equivalents and successor media, in which the software implementations herein are stored.

(177) Although the present specification describes components and functions implemented in the embodiments with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. Each of the standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same functions are considered equivalents.

(178) FIG. **34** is an illustration of a communication network **900** for measurement and reporting in accordance with an exemplary embodiment. Briefly, the communication network **900** expands broad data connectivity to other devices or services. As illustrated, the measurement and reporting

system **902** can be communicatively coupled to the communications network **900** and any associated systems or services.

(179) As one example, measurement system **902** can share its parameters of interest (e.g., angles, load, balance, distance, alignment, displacement, movement, rotation, and acceleration) with remote services or providers, for instance, to analyze or report on surgical status or outcome. This data can be shared for example with a service provider to monitor progress or with plan administrators for surgical monitoring purposes or efficacy studies. The communication network **900** can further be tied to an Electronic Medical Records (EMR) system to implement health information technology practices. In other embodiments, the communication network **900** can be communicatively coupled to HIS Hospital Information System, HIT Hospital Information Technology and HIM Hospital Information Management, EHR Electronic Health Record, CPOE Computerized Physician Order Entry, and CDSS Computerized Decision Support Systems. This provides the ability of different information technology systems and software applications to communicate, to exchange data accurately, effectively, and consistently, and to use the exchanged data.

(180) The communications network **900** can provide wired or wireless connectivity over a Local Area Network (LAN) **904**, a Wireless Local Area Network (WLAN) **910**, a Cellular Network **906**, and/or other radio frequency (RF) system (see FIG. 4). The LAN **904** and WLAN **910** can be communicatively coupled to the Internet **908**, for example, through a central office. The central office can house common network switching equipment for distributing telecommunication services. Telecommunication services can include traditional POTS (Plain Old Telephone Service) and broadband services such as cable, HDTV, DSL, VoIP (Voice over Internet Protocol), IPTV (Internet Protocol Television), Internet services, and so on.

(181) The communication network **900** can utilize common computing and communications technologies to support circuit-switched and/or packet-switched communications. Each of the standards for Internet **908** and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP, RTP, MMS, SMS) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same functions are considered equivalent.

(182) The cellular network **906** can support voice and data services over a number of access technologies such as GSM-GPRS, EDGE, CDMA, UMTS, WiMAX, 2G, 3G, WAP, software defined radio (SDR), and other known technologies. The cellular network **906** can be coupled to base receiver **912** under a frequency-reuse plan for communicating with mobile devices **914**.

(183) The base receiver **912**, in turn, can connect the mobile device **914** to the Internet **908** over a packet switched link. The internet **908** can support application services and service layers for distributing data from the measurement system **902** to the mobile device **914**. Mobile device **914** can also connect to other communication devices through the Internet **908** using a wireless communication channel. The mobile device **914** can also connect to the Internet **908** over the WLAN **910**. Wireless Local Access Networks (WLANs) provide wireless access within a local geographical area. WLANs are typically composed of a cluster of Access Points (APs) **916** also known as base stations. The measurement system **902** can communicate with other WLAN stations such as laptop **918** within the base station area. In typical WLAN implementations, the physical layer uses a variety of technologies such as 802.11b or 802.11g WLAN technologies. The physical layer may use infrared, frequency hopping spread spectrum in the 2.4 GHz Band, direct sequence spread spectrum in the 2.4 GHz Band, or other access technologies, for example, in the 5.8 GHz ISM band or higher ISM bands (e.g., 24 GHz, etcetera).

(184) By way of the communication network **900**, the measurement system **902** can establish connections with a remote server **920** on the network and with other mobile devices for exchanging data. The remote server **920** can have access to a database **922** that is stored locally or remotely and

which can contain application specific data. The remote server **920** can also host application services directly, or over the internet **908**.

(185) FIG. **35** is a diagram of a robot **1000** supporting installation of a shoulder joint in accordance with an example embodiment. In general, a robot can support or assist in the installation of the shoulder joint under control of a surgeon. In the example embodiment, measurement device **154** can be coupled to robot **1000**. One example of the robot is the Robodoc surgical robot with a robotic assisted joint installation application. Robot **1000** can also include surgical CNC robots, surgical haptic robots, surgical tele-operative robots, surgical hand-held robots, or any other surgical robot. Measurement device **154** can be automated to couple to and work with robot **1000** thereby replacing direct hand control by the surgeon. The actions taken by robot **1000** in control of measurement device **154** can be smoother and more accurate by having robot **1000** use the measurement data in real-time and providing feedback to measurement device **154** for subsequent steps. An added benefit can be shortening the time of surgery that reduces the time a patient is under anesthesia.

(186) Robot **1000** can be configured to perform computer-assisted surgery and more specifically shoulder surgery with measurement device **154**. Typically, robot **1000** and measurement device **154** is used for computer-assisted surgery to improve performance, alignment, stability, range of motion, reduce surgical time, and minimize impingement in the installation of a prosthetic joint and more specifically a shoulder joint. In one embodiment, robot **1000** can distract, perform bone cuts, align prosthetic components, reposition prosthetic components, adjust loading, perform tissue releases, perform range of motion, improve stability using the real-time measurement data sent from measurement device **154**.

(187) In general, measurement data from measurement device **154** can be wirelessly transmitted to a computer of robot **1000**. Alternatively, the measurement data can be hard wired to robot **1000**. Examples of measurement data from measurement device **154** can be range of motion for predetermined movements, impingement, load magnitude, position of load, position, and motion to name but a few. The measurement data received by robot **1000** can be further processed to calculate and display measurement data needed by the surgeon for the preparation of the bone surfaces or installation of the final prosthetic components based on the quantitative measurement data. The prepared bone surfaces will receive a prosthetic component that supports proper alignment for optimal range of motion and stability. In one embodiment, the computer in robot **1000** includes one or more algorithms that are used at various stages of the surgery. The measurement data from measurement device **154** is input to the algorithms of robot **1000** and the algorithms can convert the data into information displayed on the display for robotic actions that are used to make bone cuts, pin placements, prosthetic component sizing, etcetera or provide feedback on actions that the surgeon may take. The feedback may take the form of audible, visual, or haptic feedback that guides the surgeon on the distraction or subsequent steps taken by the robot to support or resist an action based on the measurement data. The feedback can also smooth or prevent motions by a user that could be detrimental to the surgery. Furthermore, the status of the measurement data can be used to generate a workflow that is subsequently implemented by a surgeon or automatically by robot **1000** to enhance performance and reliability of the shoulder joint installation.

(188) FIG. **36** is an illustration of a measurement device **1100** in accordance with an example embodiment. Measurement device **1100** houses electronic circuitry and at least one sensor identical to measurement device **154** as shown in FIGS. **3-21**. Measurement device **1100** transmits measurement data to a computer in proximity to display measurement data in real-time.

Measurement data from measurement device **1100** can be provided on a GUI **380** as shown in FIGS. **22A-FIG. 27A** as disclosed herein above. In the example, measurement device **1100** is configured for a shoulder implant. In general, measurement device **1100** can be adapted for use in the musculoskeletal system such as bones, tissue, ligaments, tendons, or joints. As shown, measurement device **1100** is configured to couple to a humeral prosthesis for measuring range of

motion, stability, impingement, load, and position of load of the shoulder joint. A glenoid sphere of the shoulder joint is configured to couple to an external curved surface **1104** of measurement device **1100**. Moreover, embodiments or uses stated for measurement device **154** herein above can be applied to measurement device **1100**.

(189) Measurement device **1100** comprises an upper housing **1106** and a bottom housing **1108**. Upper housing **1106** and bottom housing **1108** couple together to form a hermetically sealed enclosure that houses the electronic circuitry, power source, and sensors. Upper housing **1106** has a rim **1102** and external curved surface **1104**. Measurement device **1100** further includes a shim **1110** configured to couple to bottom housing **1108**. Shim **1110** is a removable structure of measurement device **1110** that is configured to couple to a humeral tray of the humeral prosthesis. A plurality of shims are provided with measurement device **1100** to change a height of measurement device **1100**. In one embodiment, increasing the height of measurement device **1100** using a shim can be used to increase loading applied by the muscles, tendons, or ligaments of the shoulder joint. Conversely, decreasing the height of the measurement device **1100** using a shim of a lesser height can decrease loading applied by the muscles, tendons, or ligaments of the shoulder joint. In one embodiment, a plurality of shims can be provided that change an angle that measurement device **1100** presents to the glenoid sphere when coupled in the shoulder joint.

(190) FIG. 37A is a superior view of measurement device **1100** in accordance with an example embodiment. As mentioned previously, shim **1110** removably couples to bottom housing **1108**. This allows different shims to be rapidly coupled to or removed from measurement device **1100** during a shoulder joint installation to determine an optimal fit or placement of the humeral prosthesis in the shoulder joint using quantitative measurement data. A cut out **1112** is shown in bottom housing **1108**. In one embodiment, a ledge or protrusion is formed on bottom housing **1108** that underlies retaining feature **1114**. Retaining feature **1114** is a tab extending from shim **1110**. In one embodiment, retaining feature **1114** couples to the ledge or protrusion of bottom housing **1108** under force to retain shim **1110** to bottom housing **1108**. In one embodiment, retaining feature **1114** is flexible such that retaining feature **1114** can be forced away from the ledge or protrusion of bottom housing **1108** to release shim **1110** from bottom housing **1108**. Cutout **1112** in bottom housing **1108** allows access to facilitate flexing of retaining feature **1114** away from bottom housing **1108**.

(191) FIG. 37B is a view of measurement device **1100** illustrating external curved surface **1104** in accordance with an example embodiment. In general, upper housing **1106** of measurement device **1100** includes rim **1102** and external curved surface **1104**. As mentioned previously, a glenoid sphere couples of the shoulder joint couples to external curved surface **1104** of measurement device **1100**. In one embodiment, the plurality of sensors underlie external curved surface **1104** and are configured to measure a force, pressure, or load applied at each location where a sensor is placed. The measurement data from the plurality of sensors is provided to a computer to calculate a load magnitude applied to external curved surface **1104** by the glenoid sphere of the shoulder joint and the location of the applied load by the glenoid sphere.

(192) FIG. 37C is a side view of measurement device **1100** in accordance with an example embodiment. Upper housing **1106** is coupled to bottom housing **1108**. Shim **1110** is coupled to bottom housing **1108**. The side view illustrates under-cut **1116** and under-cut **1118** in shim **1110**. In one embodiment, under-cut **1118** is also on the opposing side of shim **1110**. Under-cuts **1116** and **1118** are used to retain and align measurement device **1100** to an implant tray of the humeral prosthesis. In one embodiment, corresponding features formed in the implant tray couple to under-cuts **1116** and **1118**.

(193) FIG. 37D is an anterior view of measurement device **1100** illustrating under-cut **1118** formed in shim **1110** in accordance with an example embodiment. Under-cut **1118** is used to retain and align measurement device **1110** to the implant tray of the humeral prosthesis.

(194) FIG. 38 is an exploded view of measurement device **1100** in accordance with an example

embodiment. A flexible interconnect **1120** is configured to couple a plurality of sensors to electronic circuitry **1130**. The plurality of sensors can be formed in or on flexible interconnect **1120**. In one embodiment, the plurality of sensors comprises a sensor **1122**, a sensor **1124**, and a sensor **1126**. In one embodiment, sensors **1122**, **1124**, and **1126** are formed in flexible interconnect **1120**. Alternatively, sensors can be coupled to flexible interconnect **1120**. In one embodiment, one or more reference sensors are formed in flexible interconnect **1120**. In one embodiment, sensors **1122**, **1124**, and **1126** and interconnect in flexible interconnect **1120** are shielded. Sensors **1122**, **1124**, and **1126** are configured to couple to external curved surface **1104**.

(195) Electronic components and power source **1132** are coupled to a printed circuit board **1128**. Printed circuit board **1128** includes one or more levels of interconnect to connect the electronic components to form electronic circuitry **1130** that is configured to control a measurement process and transmit measurement data. In one embodiment, power source **1132** comprises batteries for powering measurement device **1100**. Printed circuit board **1128** can be a rigid printed circuit board that includes a connector for coupling to flexible interconnect **1120**. Electronic circuitry **1130**, flexible interconnect **1120**, and sensors **1122**, **1124**, and **1126** are placed in a cavity **1136** of bottom housing **1108**. Upper housing **1106** couples to bottom housing **1108** to form a housing for electronic circuitry **1130**, flexible interconnect **1120**, and sensors **1122**, **1124**, and **1126**. In one embodiment, sensors **1122**, **1124**, and **1126** are placed at predetermined locations between upper housing **1106** and bottom housing **1108** to support a load magnitude and position of applied load measurement where a glenoid sphere couples to external curved surface **1104** of measurement device **1100** for the shoulder joint. As mentioned previously, the electronic circuitry **1130**, the reference sensor, a position measurement system (e.g. IMU), and sensors **1122**, **1124**, and **1126** operate similarly to that described for measurement device **154** (see FIG. 3) and will not be disclosed in detail for brevity. Furthermore, measurement device **1100** will transmit measurement data to computer **162** and have measurement data displayed on display **164** as disclosed herein above. The measurement data can be displayed on GUI **380** in graphical form to speed assimilation of the measurement data in the same manner as discussed for measurement device **154**.

(196) A shim **1110** couples to bottom housing **1108** to add height to measurement device **1100**. A plurality of shims are provided with measurement device **1100** where each shim of the plurality of shims has a different thickness. Thus, shim **1110** can be removed and replaced with one of the other shims from the plurality of shims to change the height of measurement device **1100** by a predetermined amount. Upper housing **1106**, bottom housing **1108**, and shim **1110** can be formed from a biological compatible material such as a composite material, a polymer, plastic, metal, or a metal alloy. In one embodiment, upper housing **1106**, bottom housing **1108**, and shim **1110** can be molded or 3D printed from a polymer material.

(197) An implant tray **1134** is a component of the humeral prosthesis. In one embodiment, implant tray **1134** couples to the humeral prosthesis. In one embodiment, implant tray **1134** is held in place to the humeral prosthesis by a screw and implant tray **1134** can be removed by removing the screw. Shim **1110** is configured to couple to implant tray **1134** to hold measurement device **1100** in place for generating quantitative measurement data related to the shoulder joint for assessing range of motion, stability, impingement, movement, load, or position of load.

(198) FIG. 39 is a view of cavity **1136** of bottom housing **1108** of measurement device **1100** in accordance with an example embodiment. In the example, placement of components of measurement device **1100** are illustrated. Printed circuit board **1128** with the electronic components is coupled to an interior surface of bottom housing **1108**. Printed circuit board **1148** fits within cavity **1136** and is retained by printed circuit board snaps **1142**. Printed circuit board snaps **1142** are forced through openings in printed circuit board **1128** to form an interference fit that prevents movement and aligns printed circuit board **1128** within cavity **1136**. In one embodiment, printed circuit board snaps **1142** extend from the interior surface of bottom housing **1108** and the heads of printed circuit board snaps **1142** are made larger than the openings in printed circuit board **1128**.

Printed circuit board **1128** has a connector **1146** that couples to flexible interconnect **1120**. In one embodiment, power source **1132** couples to and is retained by printed circuit board **1128**. Power source **1132** provides power for electronic circuitry **1130** and sensors **1122**, **1124**, and **1126** to provide measurement data for a complete shoulder replacement surgery.

(199) Sensors **1122**, **1124**, and **1126** couple to a surface **1144** on bottom housing **1108** at predetermined locations relative to external surface **1104** of upper housing **1106** (see FIG. 36). In one embodiment, sensors **1122**, **1124**, and **1126** are located on a radial position of external curved surface **1106**. In one embodiment, sensors **1122**, **1124**, and **1126** are spaced equidistant from one another. In one embodiment, a reference sensor **1148** can be located centrally on or in flexible interconnect **1120** relative to sensors **1122**, **1124**, and **1126**. Alternatively, more than one reference sensor can be formed on or in flexible interconnect **1120**. Sensors **1122**, **1124**, and **1126** are held in place by sensor snaps **1140**. Sensor snaps **1140** couple through openings in proximity to sensors **1122**, **1124**, and **1126** to align and retain sensors **1122**, **1124**, and **1126** at the predetermined locations. In one embodiment, a raised region is formed underlying sensors **1122**, **1124**, and **1126**. The raised regions extend above surface **1144** and provide a planar surface to support sensors **1122**, **1124**, and **1126**.

(200) FIG. 40 is a cross-sectional view of measurement device **1100** in accordance with an example embodiment. Upper housing **1104** is shown coupling to bottom housing **1108**. In one embodiment, a peripheral groove **1152** is formed circumferentially on bottom housing **1108**. A peripheral tongue **1150** is formed circumferentially on upper housing **1106** and is configured to couple to peripheral groove **1152** on bottom housing **1108**. In one embodiment, glue or an adhesive can be used in peripheral groove **1152** to seal and retain upper housing **1106** to bottom housing **1108**. Alternatively, one or more retaining structures on upper housing **1106** and one or more corresponding retaining structures on bottom housing **1108** can be used to couple upper housing **1106** to bottom housing **1108**. Peripheral tongue **1150** can comprise a conformal material that forms a seal when upper housing **1106** couples to bottom housing **1108**.

(201) Retaining feature **1114** on a superior side of measurement device **1100** is shown coupling shim **1110** to bottom housing **1108**. Although not shown there can be more than one retaining feature coupling shim **1110** to bottom housing **1110**. Bottom housing **1108** has a cutout **1156** configured to receive retaining feature **1114**. Retaining feature **1114** has a corresponding protrusion **1154** configured to fit in cutout **1156**. As mentioned previously, retaining feature **1114** is flexible and can be flexed away from bottom housing **1108** such that protrusion **1154** is outside cutout **1156** allowing shim **1110** to be removed from bottom housing **1108**.

(202) Printed circuit board **1130** is retained to or in proximity to the interior surface of bottom housing **1108**. Flexible interconnect **1120** is shown coupling to connector **1146** on printed circuit board **1130**. Sensor **1122** formed in or placed on flexible interconnect is coupled between upper housing **1106** and bottom housing **1108**. In one embodiment, sensor **1122** couples to planar surfaces formed on an interior surface of upper housing **1106** and the interior surface of bottom housing **1108**. Sensor **1122** underlies a predetermined location of external surface **1104**. The predetermined locations of sensors **1122**, **1124**, and **1126** are used to calculate a position of applied load and a load magnitude from the measurement data. Note that flexible interconnect **1120** does not undergo any bends that kink the interconnect. Sensor snap **1140** is shown coupling through flexible interconnect **1120** to retain sensor **1120** at the predetermined location.

(203) FIGS. 41A and 41B illustrates measurement device **1100** with two different shims in accordance with an example embodiment. In general, measurement device **1100** includes a plurality of shims. In the example, two different shims are disclosed but more than two can be provided. Shim **1110** is a zero height shim and corresponds to measurement device **1100** at a minimum height. In the example, upper housing **1106** and bottom housing **1108** are coupled together and configured to measure at least one parameter. Shim **1110** is a separate component that couples to bottom housing **1108**. Measurement device **1100** is shown with shim **1110** coupled to bottom

housing **1110** to form measurement device **1110** at a standard, normal or minimum height.

(204) Shim **1160** is a 2.5 millimeter shim that raises the height of measurement device **1100** 2.5 millimeters when compared to zero height shim **1100** of FIG. **41A**. In the example, upper housing **1106** and bottom housing **1108** are coupled together and configured to measure at least one parameter. Shim **1160** is a separate component that couples to bottom housing **1108**. Measurement device **1100** is shown with shim **1160** coupled to bottom housing **1110** to form measurement device **1110** at a height increased by 2.5 millimeters. As stated previously, measurement device **1100** can be provided with more than two shims of different heights. For example, the shoulder joint when reduced with shim **1110** of FIG. **41A** might have a loading when measured by measurement device **1100** that is less than desirable. Shim **1110** can then be removed and replaced with shim **1160**. Reducing the shoulder joint with shim **1160** in measurement device **1160** will increase tension on muscles of the shoulder joint thereby increasing the loading applied to measurement device **1100**. The tension on the different muscles, ligaments, or tendons can be adjusted to achieve stability, maximize range of motion, minimize impingement, and load the shoulder joint within an acceptable range based on quantitative measurement data. For example, the tension can be adjusted using soft tissue tensioning to adjust the loading within the acceptable range measured by load sensors within measurement device **1100** in real-time.

(205) FIG. **42** is a cross-sectional view of external curved surface **1104** of upper housing **1106** that is modified to direct loading to predetermined areas of external curved surface **1104** in accordance with an example embodiment. The cross-sectional view shows upper housing **1106** coupled to bottom housing **1108** to form an enclosure to isolate and hermetically seal electronic circuitry **1130**, flexible interconnect, **1120**, and sensors **1122**, **1124**, and **1126** from an external environment. Shim **1110** couples to bottom housing **1108**. A partial view of glenoid sphere **1194** is shown coupling to external curved surface **1104** of upper housing **1106**. In general, loading by glenoid sphere **1194** is directed to the plurality of load sensors underlying external curved surface **1104**. In the example, the loading is directed to sensor **1122**, sensor **1124**, and sensor **1126** (not shown in the illustration). As shown, sensors **1122** and **1124** respectively underlie region **1172** and region **1174** of external curved surface of **1104**. Similarly, sensor **1126** will underlie region **1176** of FIG. **42**. As previously mentioned, sensors **1122**, **1124**, and **1126** are located at predetermined radial positions of external curved surface **1104**. In one embodiment, sensors **1122**, **1124**, and **1126** are also spaced equidistant from each other. Sensors **1122**, **1124**, and **1126** are placed as close to rim **1102** of upper housing **1106** as feasible to maximize the measurement area on external curved surface **1104**.

(206) In the example, glenoid sphere **1194** is 38 millimeters in diameter and has a 19 millimeter radius. In one embodiment, external curved surface **1104** has a larger radius than glenoid sphere **1194**. In the example, the radius of external curved surface is 38.15 millimeters. External curved surface **1104** is modified such that glenoid sphere **1194** only couples to regions **1172**, **1174**, and **1176** of external curved surface **1104**. Sensors **1122**, **1124**, and **1126** respectively underlie regions **1172**, **1174**, and **1176** of external curved surface **1104**. Thus, loading applied by glenoid sphere **1194** is directed to sensors **1122**, **1124**, and **1126** of measurement device **1100** and not to areas of external curved surface **1104** outside regions **1172**, **1174**, and **1176**. In general, there are two regions on external curved surface **1104** that does not couple to glenoid sphere **1194**. In one embodiment, a region **1192** of external curved surface **1104** of upper housing **1106** does not couple to glenoid sphere **1194**. Note that a gap is shown between glenoid sphere **1194** and external curved surface **1104** in region **1192**. Region **1192** corresponds to a load measurement area between sensors **1122**, **1124**, and **1126**. In one embodiment, the gap between glenoid sphere **1194** and external curved surface **1104** in region **1192** is approximately 0.15 millimeters. In one embodiment, region **1192** can be molded having a 0.15 millimeter cutout in region **1192**. Alternatively, 0.15 millimeter of material can be removed from region **1192**.

(207) In one embodiment, a region **1190** of external curved surface **1104** of upper housing **1106** does not couple to glenoid sphere **1194**. Note that a gap is shown between glenoid sphere **1194** and

external curved surface **1104** in region **1130**. Region **1190** corresponds to the area outside region **1192** and regions **1172**, **1174**, and **1176** of external curved surface **1104**. In one embodiment, the gap between glenoid sphere **1194** and external curved surface **1104** in region **1190** is approximately 0.10 millimeters. In one embodiment, can be molded having a 0.10 millimeter cutout in region **1190**. Alternatively, 0.10 millimeter of material can be removed from region **1190**.

(208) FIG. **43** is a block diagram of loading measurement device **1100** in accordance with an example embodiment. The method will refer to components listed in FIG. **42**. In a step **1180**, a glenoid sphere **1194** of a shoulder joint loads measurement device **1100**. In the example, the glenoid sphere **1194** couples to the scapula and measurement device **1100** couples to the humeral prosthesis to form the shoulder joint. In a step **1182**, glenoid sphere **1194** couples to regions **1172**, **1174**, and **1176** directly. Sensors **1122**, **1124**, and **1126** respectively underlie regions **1172**, **1174**, and **1176** of external curved surface **1104** of upper housing **1106**. In one embodiment, glenoid sphere **1194** does not couple to regions **1190** and **1192** of external curved surface **1104** of upper housing **1106**. Loading applied by glenoid sphere **1194** is coupled through and distributed among sensors **1122**, **1124**, and **1126**. In a step **1186**, by distributing the load through regions **1172**, **1174**, and **1176** and thereby through sensors **1122**, **1124**, and **1126** the expected sensitivity of the sensors is correct. In one embodiment, loading applied by glenoid sphere **1194** to measurement device **1100** only couples through sensors **1122**, **1124**, and **1126**.

(209) FIG. **44** is an illustration of measurement device **1100** illustrating different regions of external curved surface **1104** of upper housing **1106** in accordance with an example embodiment. In general, external curved surface **1104** has three different regions each having a different surface height. The first regions corresponds to locations of the sensors for measuring a force, pressure, or load applied to external curved surface **1104**. The surface height and curvature of the first regions corresponds to a radius configured to receive a spherical prosthetic component. In the example, the glenoid sphere has a radius of 19 millimeters and the radius of the first regions corresponding to the locations of the sensors is 19.075 millimeters. In one embodiment, external curved surface **1104** has a larger radius than spherical prosthetic component that couples to it.

(210) The second region of external curved surface **1104** corresponds to a location of applied load of the spherical prosthetic component to external curved surface **1104**. In general, the sensors are located adjacent to or in proximity to rim **1102** of upper housing **1106**. Placing the sensors in proximity to the rim **1102** maximizes area in which the sensors can accurately measure the location of applied load. In one embodiment, the sensors are located on a radial position of external curved surface **1104**. In one embodiment, the sensors are spaced equidistant from one another. In one embodiment, the second region can be located at or below the sensor locations. In one embodiment, the second region can be an irregular shape. In one embodiment, the second region can comprise more than one second region. In the example, the surface of the second region of external curved surface **1104** is below the surface of the first regions. In one embodiment, the spherical prosthetic component does not couple to the second region when coupled to external curved surface **1104**. The spherical prosthetic component couples to the first regions corresponding to the sensor locations.

(211) The third region of external curved surface **1104** corresponds to a location of applied load that is in proximity to the sensors or above the sensors on external curved surface **1104**. In one embodiment, the range of motion of the spherical prosthetic component when coupled to external curved surface **1104** does not typically place the position of applied load near rim **1102** of upper housing **1106**. In general, the third region corresponds to extremes of the range of motion for the prosthetic joint. In one embodiment, the third region can be located in proximity to or above the sensor locations. In one embodiment, the third region can be an irregular shape. In one embodiment, the third region can comprise more than one third region. In the example, the third region of external curved surface **1104** is outside the second region of external curved surface **1104** but does not include the first regions. In one embodiment, the spherical prosthetic component does

not couple to the third region when coupled to external curved surface **1104**. The spherical prosthetic component couples to the first regions corresponding to the sensor locations. In the example, the surface of the third region is below the surface of the first regions. In one embodiment, the surface of the second region is below the surface of the third region.

(212) In the example, a circle **1196** is drawn on external curved surface **1104** to define a boundary that identifies a region **1190** and a region **1192**. Region **1192** corresponds to the second region of external curved surface **1104** disclosed herein above. Region **1192** is an area of external curved surface **1104** that is within circle **1196**. Region **1190** corresponds to the third region of external curved surface **1104**. In one embodiment, region **1190** is an area of external curved surface **1104** that is outside of circle **1196** but does not include regions **1172**, **1174**, and **1176** up to rim **1102**. As mentioned previously, sensors **1122**, **1124**, and **1126** as shown in FIG. **39** respectively underlie regions **1172**, **1174**, and **1176**. In one embodiment, regions **1172**, **1174**, and **1176** have an area larger than or equal to an area of sensors **1122**, **1124**, and **1126**. In one embodiment, regions **1172**, **1174**, and **1176** have a curved surface configured to interface with glenoid sphere **1194** of FIG. **42**. In the example, regions **1172**, **1174**, and **1176** have a curved surface corresponding to a radius of 19.075 millimeters whereas the glenoid sphere has a radius of 19 millimeters. A surface of region **1192** of external curved surface **1104** is below the surface of regions **1172**, **1174**, and **1176**. In the example, the surface of region **1192** is 0.15 millimeters below the surfaces of regions **1172**, **1174**, and **1176** such that glenoid sphere **1194** of FIG. **42** does not couple to region **1192** when glenoid sphere **1194** is coupled to measurement device **1100**. A surface of region **1190** of external curved surface **1104** is below the surface of regions **1172**, **1174**, and **1176**. In the example, the surface of region **1190** is 0.10 millimeters below the surfaces of regions **1172**, **1174**, and **1176** such that glenoid sphere **1194** does not couple to region **1190** when glenoid sphere **1194** is coupled to measurement device **1100**. In the example, the surface of region **1190** is above the surface of region **1192** in relation to regions **1172**, **1174**, and **1176**. In one embodiment, having regions **1190**, **1192**, **1172**, **1174**, and **1176** of external curved surface **1104** directs loading applied by glenoid sphere **1194** through sensors **1122**, **1124**, and **1126** which yields an expected sensitivity for the measurement system.

(213) Referring briefly to FIGS. **1-3**, a shoulder joint system **160** is disclosed comprising a first shoulder prosthesis and a second shoulder prosthesis. In general, shoulder joint system **160** can be used for a prosthetic reverse shoulder or a normal prosthetic shoulder. The electronic circuitry and sensors can be housed in either a prosthetic component that couples to the humerus, a prosthetic component that couples to the scapula, or both. In one embodiment, the first shoulder prosthesis is a humeral prosthesis **158** configured to couple to humerus **150**. Humeral prosthesis **158** comprises a stem **124**, a neck **126**, and a tray **156**. In one embodiment, the second shoulder prosthesis is a glenoid sphere **152** configured to couple to scapula **140**. Humeral prosthesis **158** and glenoid sphere **152** each have external curved surfaces configured to mate together to support movement and rotation of the shoulder joint. Alternatively in normal prosthetic shoulder, the first shoulder prosthesis is a glenoid prosthesis **114** and the second shoulder prosthesis is a humeral prosthesis **102**.

(214) During a trialing process a humeral liner **128** of humeral prosthesis **158** is removed and a measurement device **154** replaces humeral liner **128** to take one or more measurements to support installation of shoulder joint system **160**. Measurement device **154** has an external curved surface configured to mate with glenoid sphere **152** and support movement of the shoulder joint system. In one embodiment, measurement device will be dimensionally identical to the humeral liner that couples to humeral tray **156**. Referring briefly to FIGS. **5-21**, electronic circuitry and a plurality sensors are shown in measurement device **154**. In particular, FIGS. **5** and **6** illustrate measurement device **154** having an upper housing **220** and a bottom housing **222**. Three sensors **230** are illustrated that are located at different radial locations beneath external curved surface **224** of upper housing **220**. Upper housing **220** couples to bottom housing **222** to form a hermetic seal that houses

the electronic circuitry and a plurality of sensors. In one embodiment, the hermetic seal is formed by an o-ring or adhesive that couples between upper housing **220** and bottom housing **222**. Referring briefly to FIG. **16**, one or more housing snaps **279** are used to couple upper housing **220** to bottom housing **222** of measurement device **154**. In one embodiment, a housing snap **278** comprises a protrusions formed on a sidewall **328** of bottom housing **222** and a corresponding opening formed on upper housing **220**. The region of upper housing **220** having the openings is configured to flex such that the upper housing **220** can be forcibly pressed onto bottom housing **222** until the opening overlies a corresponding housing snap **278**. The one or more housing snaps **278** preloads the plurality of sensors when coupling upper housing **220** to bottom housing **222**. Referring briefly to FIG. **29B** a plurality of sensors are each located a radial position relative to the external curved surface of measurement device **154**. In one embodiment 3 sensors are located at 3 different radial locations. Referring briefly to FIG. **31**, force sensors **530**, **532**, and **534** are oriented such that the reaction forces are directed to a center of rotation of a shoulder prosthesis. A block diagram of electronic circuitry **236** of measurement device **154** is disclosed in FIG. **32**. Electronic circuitry **236** operatively couples to the plurality of sensors in measurement device **154**. Electronic circuitry **236** controls a measurement process and transmits measurement data.

(215) Referring briefly to FIG. **19**, a flexible interconnect **228** couples the plurality of sensors **230** to electronic circuitry **236**. Referring briefly to FIG. **14**, flexible interconnect **228** couples a support structure **262** to place plurality of sensors **230** in position relative to the external curved surface of upper housing **220**. In one embodiment, flexible interconnect **228** couples to an interior surface **244** of bottom housing **222**. In one embodiment, flexible interconnect **228** couples to support structure **262** a predetermined angle such that the surface of support structure **262** is non-parallel to interior surface **244** of bottom housing **222**. In one embodiment, support structure **262** positions plurality of sensor **230** to couple between upper housing **220** and bottom housing **222**. Each sensor of plurality of sensors **230** underlies the external curved surface of upper housing **220** at predetermined locations discussed herein. Each sensor of plurality of sensors **230** respectively couples to a planar surface on the interior of the upper housing and a planar surface of the bottom housing. In one embodiment, the three load sensors are placed equidistant from each other.

(216) Electronic circuitry **236** and the plurality of sensors transmit measurement data to a computer **162** of FIG. **32**. Computer **162** is configured to receive the measurement data from the shoulder joint system that includes measurement device **154** but can include sensors in other components of the shoulder joint system. Electronic circuitry **236** of the first shoulder prosthesis and computer **162** can be in two-way communication. In one embodiment, three load sensors are configured to measure loading applied to the first shoulder prosthesis. In one embodiment, computer **162** is configured to calculate a force magnitude and location of applied force using the measurements and locations of the first, second, and third sensors underlying the external curved surface of measurement device **154**. In one embodiment, the force applied to the external curved surface of measurement device **154** is normal to the external curved surface. In one embodiment, a display couples to computer **162**. The display is configured to show in real-time at least one of a load magnitude applied to the external curved surface of the first shoulder prosthesis. In general, a force applied by the second shoulder prosthesis to the first shoulder prosthesis is normal to the external curved surface of the first shoulder prosthesis. In one embodiment, the plurality of load sensors are oriented such that the reaction forces are directed to a center or rotation.

(217) Referring briefly to FIG. **22a**, display **164** coupled to computer **162** is configured to graphically display an external curved surface **384** corresponding to the external curved surface of the upper housing. Computer **162** is configured to receive measurement data from measurement device **154**. Computer **162** can perform calculations related to the measurement data from the one or more sensors. Computer **162** further convert the measurement data to a graphic form that allows a surgeon or surgical team to rapidly assimilate measurement data. Display **164** is coupled to computer **162** having a graphical user interface (GUI) **380**. GUI **380** is configured to provide an

image of external curved surface **384** of measurement device **154** and a rim surrounding the external curved surface **384**.

(218) Display **164** is configured to graphically display a contact point **382** of the upper housing where the glenoid sphere couples to the external curved surface of the upper housing in real-time. In one embodiment, glenoid sphere **152** applies a force to measurement device **154** that is normal to external curved surface **224**. Furthermore, the display also shows a contact point on the external curved surface of the first shoulder prosthesis. The contact point will move in real-time on the display as the shoulder is moved through different range of motions. In one embodiment, the display and computer is in the operating room to provide the information in real-time to the surgical team. Other measurements that are made with shoulder joint system **160** are motion, position, joint stability, range of motion, or impingement to name but a few. Referring briefly to FIG. **22B**, computer **162** is configured to calculate when impingement occurs from the measurement data. In one embodiment, display **164** notifies when impingement occurs by highlighting a rim **520** of the external curved surface of the upper housing. Rim **520** highlights a portion of the rim where the impingement occurs or corresponds to at a terminus of direction of the range of motion where the impingement occurs.

(219) In one embodiment, one or more motion bars are displayed on display **164**. The one or more motion bars are configured to graphically indicate a range of motion of the shoulder joint system as it is moved through a predetermined motion. Referring briefly to FIG. **23**, four motion bars (**400**, **402**, **404**, and **406**) are disclosed each corresponding to a specific motion. Each motion bar graphically illustrates a range of motion achieved in the installation of the shoulder joint system for the predetermined motion. Each motion bar has a first end and a second end. The first end and the second end of a motion bar corresponds to a maximum internal rotation for a predetermined movement and a maximum external rotation for the predetermined movement. Between the first end and second end, the motion bar is configured to indicate an acceptable internal range of motion and an acceptable external range of motion for the predetermined movement. The motion bar is also configured to indicate a center of the predetermined movement. In one embodiment, a bar indicates a position of the shoulder during the predetermined movement. For example, a bar **426** of motion bar **402** indicates a position of the shoulder joint in an I/E rotation at 45 degrees adduction. A box **428** on the display indicates loading applied to the external curved surface of measurement device over the predetermined movement via a color map.

(220) Referring briefly to FIG. **25**, a range of motion (ROM) overlay **390** on GUI **380** is disclosed. The motion data and load data from the predetermined motions of FIG. **23** are stored in memory. In FIG. **25**, GUI **380** graphically displays the movement of contact point **382** to the external curved surface **384** on GUI **380** for each of the four different shoulder joint movement measured in FIG. **23**. A trace of a predetermined movement on the external curved surface of the measurement device as shown in FIG. **25** is a ROM overlay. GUI **380** is configured to provide at least one ROM overlay on the image of the external curved surface of the measurement device on the display of the computer. Stated differently a ROM overlay comprises at least one load track corresponding to a predetermined movement of the shoulder joint. In one embodiment, the applied loading can vary significantly at different points on the load track. In one embodiment, the GUI **380** is configured to provide at least one impingement ROM assessment that includes a trace that illustrates the limits of abduction/adduction and horizontal flexion.

(221) In one embodiment, GUI **380** is configured to provide an image of the external curved surface of the measurement device and a rim surrounding the external curved surface. The portion of the rim is highlighted when impingement occurs and corresponds to the direction of movement of the shoulder joint. GUI **380** is configured to provide at least one impingement ROM assessment that includes a trace that illustrates the limits of abduction/adduction and horizontal flexion.

(222) Referring briefly to FIG. **3**, a shoulder system **160** is disclosed. Shoulder system **160** comprises a first prosthetic component and a second prosthetic component. Shoulder system **160** is

configured to transmit measurement data to a computer **162** and display measurement data on a display **164**. The first prosthetic component and the second prosthetic component each have a curved surface configured to couple together to support shoulder system **160**. The curved surfaces of the first and second prosthetic component allow for a wide range of motion for the shoulder joint. In one embodiment, the first prosthetic component is a humeral prosthesis **158** configured to couple to humerus **150**. In one embodiment, the second prosthetic component is a glenoid sphere **152** configured to couple to scapula **140**. Humeral prosthesis **158** comprises a humeral tray **156**. A humeral liner or measurement device **154** is configured to couple to humeral tray **156**. Referring briefly to FIG. **36**, a measurement device **1100** is disclosed. Measurement device **1100** comprises an upper housing **1106**, a bottom housing **1108**, and a shim **1110**. Measurement device **1100** or measurement device **154** is configured to couple to humeral tray **156** for providing measurement data. Shim **1110** is a removable device to adjust a height of measurement device **1100**. For example, adding height can increase the force applied to measurement device **1100** when placed in a joint. Conversely, removing the shim or replacing the shim with a thinner shim will reduce the force applied to measurement device **1100** when placed in the joint. In one embodiment, shim **1110** couples to bottom housing **1108** and retains measurement device **1100** to humeral tray **156**. Upper housing **1106** has an external curved surface **1104** configured to couple to another prosthetic component to support joint movement. Upper housing **1106** couples to bottom housing **1108** to form a hermetically sealed enclosure. The hermetically sealed enclosure houses at least one sensor and electronic circuitry configured to control a measurement process and transmit measurement data. In one embodiment, shim **1110** is configured to couple bottom housing **1108**. In one embodiment, shim **1110** is configured to couple measurement device **1100** to a humeral tray of a humeral prosthesis. Although not shown, for measurement device **154**, shim **1110** is used to change the height of measurement device **154** to adjust the load applied to the prosthetic component. In one embodiment, measurement device **1100** with shim **1110** couples to humeral tray **156** of FIG. **3**. In one embodiment, a plurality of shims are provided where each shim has a different height for adjusting a height of measurement device **1100**. Each shim of the plurality of shims is configured to couple to measurement device **1100**. Furthermore, each shim is configured to couple to a tray of the prosthetic component. In one embodiment, each shim of the plurality of shims includes one or more cutouts to retain a shim of the plurality of shims to the tray of the prosthetic component.

(223) In general, everything disclosed for measurement device **154** herein above also applies to measurement device **1100**. In other words, electronic circuitry, sensors, or structures of measurement device **154** also apply to measurement device **1100**. Although not shown the electronic circuitry disclosed for measurement device **154** herein above is also within measurement device **1100**. Thus, figures related to measurement device **154** will be disclosed when discussing structure, electronic circuitry, or sensors for measurement device **1100**. Electronic circuitry **236** is disclosed in FIG. **32**. Electronic circuitry **236** is placed within the enclosure formed by upper housing **1106** coupling to bottom housing **1108**. Referring briefly to FIG. **19** a first sensor, a second sensor, and a third sensor are located within bottom housing **222** corresponding to bottom housing **1108**. The first, second, and third sensors correspond to sensors **230**. Referring to FIG. **5**, upper housing **220** corresponding to upper housing **1106** is shown prior to coupling to bottom housing **222**. Coupling upper housing **220** to bottom housing **222** couples the first, second, and third sensors respectively to external curved surface **224** corresponding to external curved surface **1104**. More specifically, the first, second, and third sensors respectively couple to external curved surface **224** at a first predetermined radial location, a second predetermined radial location, and a third predetermined radial location upper housing **220** when coupled to bottom housing **222**. Referring briefly to FIG. **28**, an illustration of sensors **530** and **532** are shown in a cross-sectional view illustrating radial locations. The third sensor (not shown) is spaced such that all three sensors are spaced equidistant. In other words, the first, second, and third predetermined radial locations have

an equal radius from the center of curvature. In one embodiment, the first, second, and third radial locations of sensors **530**, **532**, and **534** on external curved surface **224** has a greater radius than a radius of the external curved surface of the prosthetic component that couples to measurement device **154**. In one embodiment, the first, second, and third sensors are located at or near the rim of the external curved surface. In one embodiment, the positions of the first, second, and third sensors have an equal radius relative to the external curved surface.

(224) In general, measurement device **1100** couples to a first prosthetic component for generating measurement data related to a shoulder joint system. Measurement device **1100** of the first prosthetic component couples to an external curved surface of a second prosthetic component as disclosed in FIGS. **1-3**. In one embodiment, the external curved surface of the second prosthetic component is configured to couple to the external curved surface of upper housing **220** of the measurement device **154** only at the surfaces of the first, second, and third predetermined radial locations corresponding respectively to the locations of first, second, and third sensors underlying the external curved surface of the upper housing. A force applied by the second shoulder prosthesis to the first shoulder prosthesis is directed thru the first, second, and third sensors. As previously mentioned electronic circuitry **236** couples to the first, second, and third sensors. Electronic circuitry **236** is configured to control a measurement process and transmit measurement data.

(225) Referring briefly to FIG. **44**, measurement device **1100** is illustrated disclosing different regions of external curved surface **1104** of upper housing **1006**. The regions disclosed herein also correspond to measurement device **154**. A plurality of regions correspond to a location of external curved surface **1104** where a force, pressure, or load is measured. In one embodiment, the plurality of regions correspond to a first sensor, a second sensor, and a third sensor that underlies external curved surface **1104**. Region **1172**, region **1174**, and region **1176** correspond to the plurality of regions of external curved surface **1104** and are located at a first, second, and third predetermined radial location. In one embodiment, regions **1172**, **1174**, and **1176** correspond to surfaces of the first, second, and third predetermined radial locations of the external curved surface **1104** of measurement device **1100**. Regions **1172**, **1174**, and **1176** are equal to or larger than the area of the first, second, or third sensors that underlie the radial locations. A first region of external curved surface **1104** relates to a circle **1196** illustrated on FIG. **44**. In one embodiment, a first region is the area within circle **1196** of external curved surface **1104** of measurement device **1100**. The first region is illustrated as region **1192**. In one embodiment, the surfaces of the first, second, and third predetermined radial locations corresponding to regions **1172**, **1174**, and **1176** are above a surface of the first region (region **1192**) of external curved surface **1104** within circle **1196**. A second region of external curved surface **1104** of measurement device **1100** corresponds to a region outside circle **1196** but does not include regions **1172**, **1174**, and **1176**. The second region is illustrated as region **1190**. The second region extends from circle **1196** to rim **1102** of measurement device **1100** excluding regions **1172**, **1174**, and **1176**. In one embodiment, the surfaces of the first, second, and third predetermined radial locations corresponding to regions **1172**, **1174**, and **1176** are above a surface of the second region (region **1190**). In one embodiment, the surface of the second region of external curved surface **1104** comprises an area that is above the surface of the first region of external curved surface **1104**. In one embodiment, rim **1102** of measurement device is **1100** is a maximum height of external curved surface **1104**. Conversely, the first region corresponding to the area of the surface within circle **1196** is the lowest portion of external curved surface **1104** of measurement device **1100**.

(226) A method of measurement within a shoulder joint is provided herein below. The steps disclosed can be practiced independently and in any order. The ordering of steps does not indicate an order or sequence but is merely to identify a step. In a first step, a first shoulder prosthesis is coupled to a second shoulder prosthesis. The first shoulder prosthesis has an external curved surface configured to couple to an external curved surface of the second shoulder prosthesis. In one embodiment, the external curved surface of the first shoulder prosthesis is a measurement device

configured to control a measurement process and transmit measurement data to a computer. The computer includes a display for displaying the measurement data or to graphically display information related to the measurement data. In a second step, a force, pressure, or load of the second shoulder prosthesis is directed through a surface of a first predetermined radial location, a surface of a second predetermined radial location, and a surface of a third predetermined radial location of the first shoulder prosthesis. Underlying the surface of the first predetermined radial location, the surface of the second predetermined radial location, and the third radial location respectively is a first sensor, a second sensor, and a third sensor. In a third step, the first, second, and third sensors are placed equidistant from one another at positions that maximize the radius of circle defined by the sensors. The first, second, and third sensors are oriented such that sensor reaction forces are directed to a center of curvature of the external curved surface of the first shoulder prosthesis. In one embodiment, it is assumed that no frictional forces or negligible frictional forces occur on the external curved surface of the first shoulder prosthesis or at a sensor interface. In one embodiment, reaction force vectors are assumed to be normal to the external curved surface of the first shoulder prosthesis and therefore pass through the center of curvature of the external curved surface of the first shoulder prosthesis.

(227) It should be noted that very little data exists on implanted orthopedic devices. Most of the data is empirically obtained by analyzing orthopedic devices that have been used in a human subject or simulated use. Wear patterns, material issues, and failure mechanisms are studied. Although, information can be garnered through this type of study it does not yield substantive data about the initial installation, post-operative use, and long term use from a measurement perspective. Just as each person is different, each device installation is different having variations in initial loading, balance, and alignment. Having measured data and using the data to install an orthopedic device will greatly increase the consistency of the implant procedure thereby reducing rework and maximizing the life of the device. In at least one exemplary embodiment, the measured data can be collected to a database where it can be stored and analyzed. For example, once a relevant sample of the measured data is collected, it can be used to define optimal initial measured settings, geometries, and alignments for maximizing the life and usability of an implanted orthopedic device.

(228) The present invention is applicable to a wide range of medical and nonmedical applications including, but not limited to, frequency compensation; control of, or alarms for, physical systems; or monitoring or measuring physical parameters of interest. The level of accuracy and repeatability attainable in a highly compact measurement device or surgical apparatus may be applicable to many medical applications monitoring or measuring physiological parameters throughout the human body including, not limited to, bone density, movement, viscosity, and pressure of various fluids, localized temperature, etc. with applications in the vascular, lymph, respiratory, digestive system, muscles, bones, and joints, other soft tissue areas, and interstitial fluids.

(229) While the present invention has been described with reference to particular embodiments, those skilled in the art will recognize that many changes may be made thereto without departing from the spirit and scope of the present invention. Each of these embodiments and obvious variations thereof is contemplated as falling within the spirit and scope of the claimed invention, which is set forth in the claims. While the subject matter of the invention is described with specific examples of embodiments, the foregoing drawings and descriptions thereof depict only typical embodiments of the subject matter and are not therefore to be considered to be limiting of its scope, it is evident that many alternatives and variations will be apparent to those skilled in the art. Thus, the description of the invention is merely descriptive in nature and, thus, variations that do not depart from the gist of the invention are intended to be within the scope of the embodiments of the present invention. Such variations are not to be regarded as a departure from the spirit and scope of the present invention.

(230) While the present invention has been described with reference to embodiments, it is to be

understood that the invention is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all modifications, equivalent structures and functions. For example, if words such as “orthogonal”, “perpendicular” are used the intended meaning is “substantially orthogonal” and “substantially perpendicular” respectively. Additionally although specific numbers may be quoted in the claims, it is intended that a number close to the one stated is also within the intended scope, i.e. any stated number (e.g., 90 degrees) should be interpreted to be “about” the value of the stated number (e.g., about 90 degrees).

(231) As the claims hereinafter reflect, inventive aspects may lie in less than all features of a single foregoing disclosed embodiment. Thus, the hereinafter expressed claims are hereby expressly incorporated into this Detailed Description of the Drawings, with each claim standing on its own as a separate embodiment of an invention. Furthermore, while some embodiments described herein include some but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention, and form different embodiments, as would be understood by those skilled in the art.

## Claims

1. A system for monitoring an implantable device, the system comprising: a measurement device positioned within a joint, the measurement device configured to generate measurement data and detect impingement as the joint is moved through a set of predetermined motions; and a computer configured to wirelessly communicate with the measurement device, the computer configured to process the measurement data and display the measurement data and impingement detection on a graphical user interface (GUI), wherein the displayed data provides real-time information regarding motion of the implantable device and load on the implantable device, and wherein the GUI includes a depiction of a range of motion including a maximum internal rotation or a maximum external rotation of the joint.
2. The system of claim 1, wherein the GUI displays a representation of the measurement device, and wherein the representation includes a display of a contact point, the contact point being calculated based on the measurement data received from the measurement device.
3. The system of claim 2, wherein the GUI further includes a depiction of one or more angles based on data from one or more sensor associated with the measurement device.
4. The system of claim 2, wherein the measurement data comprises information from at least three load sensors measuring load applied to an external surface of the measurement device.
5. The system of claim 4, wherein the measurement data further comprises information from at least one reference sensor and a position measurement system configured to measure position or motion.
6. The system of claim 5, wherein the position measurement system is an inertial measurement unit (IMU) housed within the measurement device.
7. The system of claim 1, wherein the GUI further includes a display of an articulation of a humeral component relative to a glenoid component.
8. The system of claim 1, wherein the GUI provides audible or haptic notifications when detecting an impingement.
9. The system of claim 1, wherein the GUI includes a motion bar providing information on a predetermined range of motion.
10. A method for monitoring an implantable device, comprising the steps of: positioning a measurement device within a joint, the measurement device configured to generate measurement data and detect impingement as the joint is moved through a set of predetermined motions; communicating the generated measurement data and impingement detection to a computer; processing the measurement data with the computer; and displaying the processed data and

impingement detection on a graphical user interface (GUI), wherein the displayed processed data provides real-time information regarding motion of the implantable device and load on the implantable device; and displaying in the GUI a range of motion analysis including a depiction of a maximum flexion of the joint.

11. The method of claim 10, further comprising: displaying a representation of the measurement device on the GUI, where the representation includes a depiction of a contact point, the contact point being calculated based on the measurement data received from the measurement device.

12. The method of claim 11, further comprising: wherein the range of motion analysis further includes a depiction of a maximum internal rotation or a maximum external rotation of the joint.

13. The method of claim 11, wherein the measurement data comprises information from at least three load sensors measuring load applied to an external surface of the measurement device.

14. The method of claim 13, wherein the measurement data further comprises information from at least one reference sensor and a position measurement system configured to measure position or motion.

15. The method of claim 14, wherein the position measurement system is an inertial measurement unit (IMU) housed within the measurement device.

16. The method of claim 10, further comprising: displaying a plot on the GUI depicting an articulation of a humeral component relative to a glenoid component.

17. The method of claim 10, further comprising: providing audible or haptic notifications on the GUI when impingement is detected.

18. The method of claim 10, further comprising: displaying on the GUI a motion bar that provides information on a set of predetermined motions.

19. A prosthetic joint assessment system comprising: a computer configured to receive data associated with a prosthetic device configured for placement within a joint; process the data; and display a graphical user interface (GUI) that displays the processed data and impingement detection; wherein the GUI is configured to display a representation of the prosthetic device and the joint; and wherein the GUI is configured to display a depiction of a range of motion including at least one of a maximum internal rotation of the joint, a maximum external rotation of the joint, or a maximum flexion of the joint.

20. The prosthetic joint assessment system of claim 19, further comprising: a position measurement system housed in a measurement device, configured to measure position or motion; and a processor embedded in the computer or the measurement device and configured to calculate a contact point, detect impingements, and measure load magnitude from data received from the measurement device.

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